

# Data Warehousing/BI Concepts

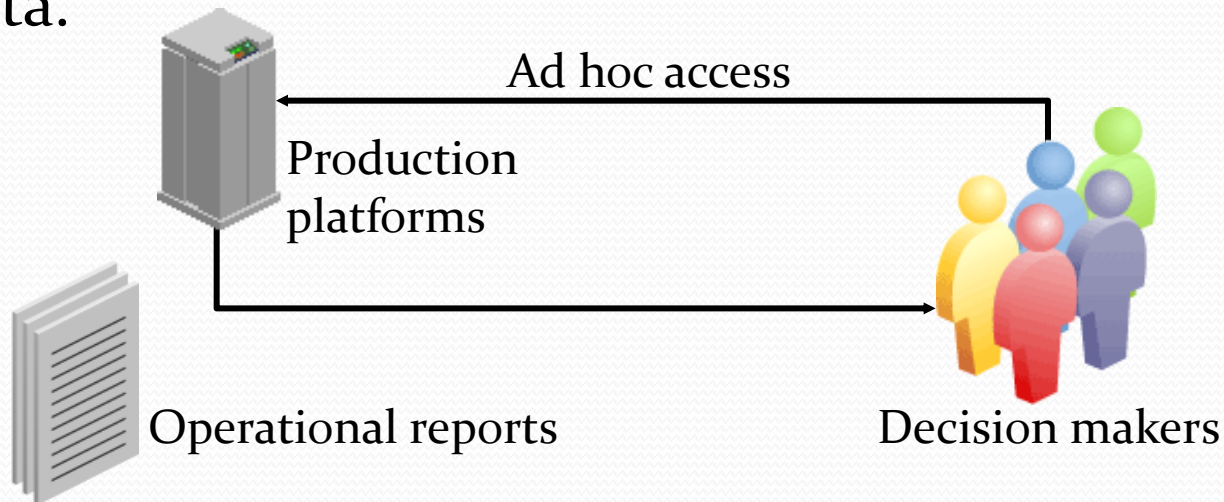
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Data Science & Big Data Analytics Track  
Intake 38

# A Brief History of BI

- The term BI was coined by the Gartner Group in the mid-1990s
- However, the concept is much older
  - 1970s — MIS reporting — static/periodic reports
  - 1980s — Executive Information Systems (EIS)
  - 1990s — OLAP, dynamic, multidimensional, ad-hoc reporting -> coining of the term “BI”
  - 2005+ — Inclusion of AI and Data/Text Mining capabilities; Web-based Portals/Dashboards
  - 2010s — BI Term is evolved to include BA & BPM

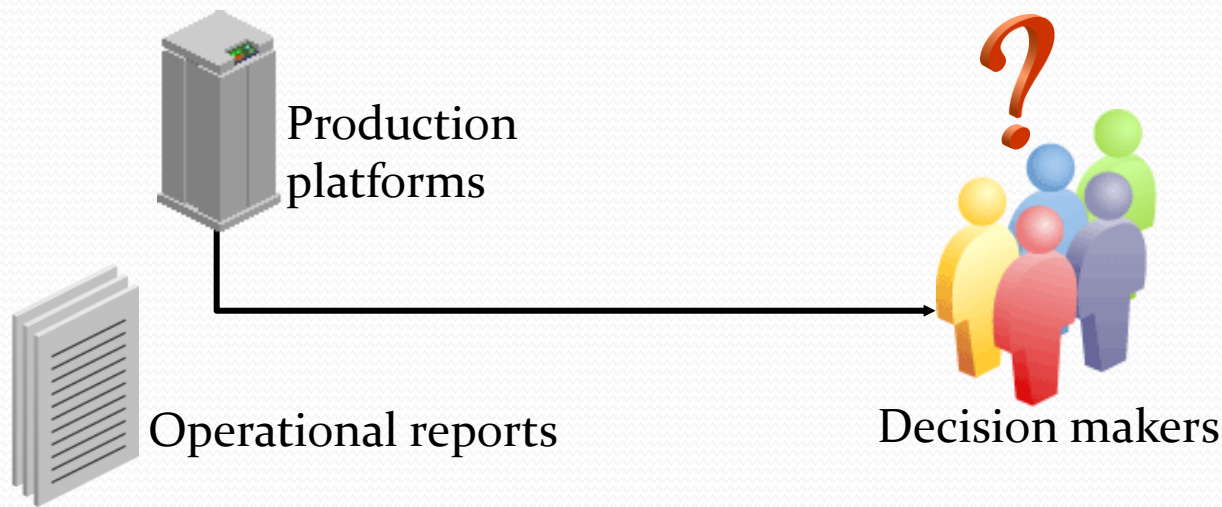
# Early Management Information Systems

- MIS systems provided business data.
- Reports were developed on request.
- Reports provided little analysis capability.
- Decision support tools gave personal ad hoc access to data.



# Analyzing Data from Operational Systems

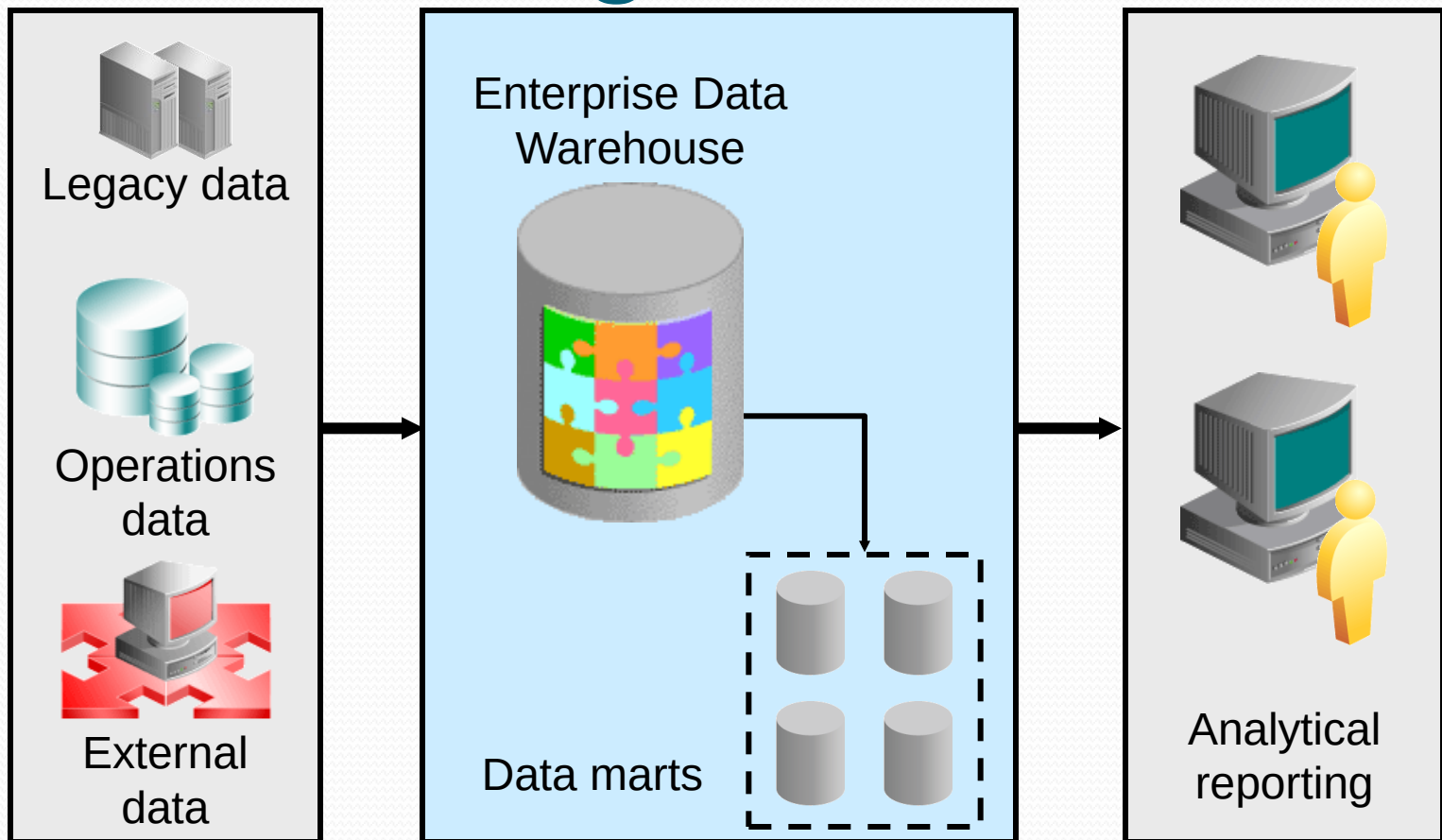
- Data structures are complex.
- Systems are designed for high performance and throughput.
- Data is not meaningfully represented.
- Data is dispersed.
- Online transaction processing (OLTP) systems may be unsuitable for intensive queries.



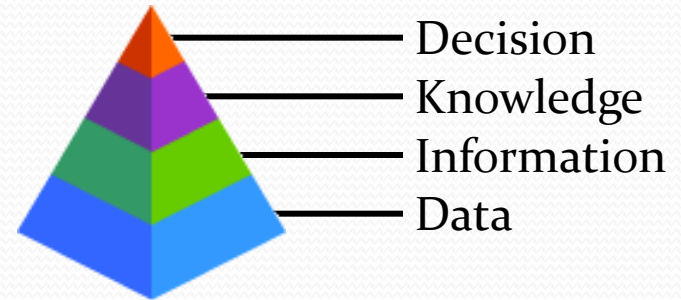
# Why Online Transaction Processing (OLTP) Is Not Suitable for Analytical Reporting

| OLTP                                      | Analytical Reporting                       |
|-------------------------------------------|--------------------------------------------|
| Information to support day-to-day service | Historical information to analyze          |
| Data stored at transaction level          | Data needs to be integrated                |
| Database design: Normalized               | Database design: Denormalized, star schema |

# Data Warehousing and Business Intelligence



# Definition of BI



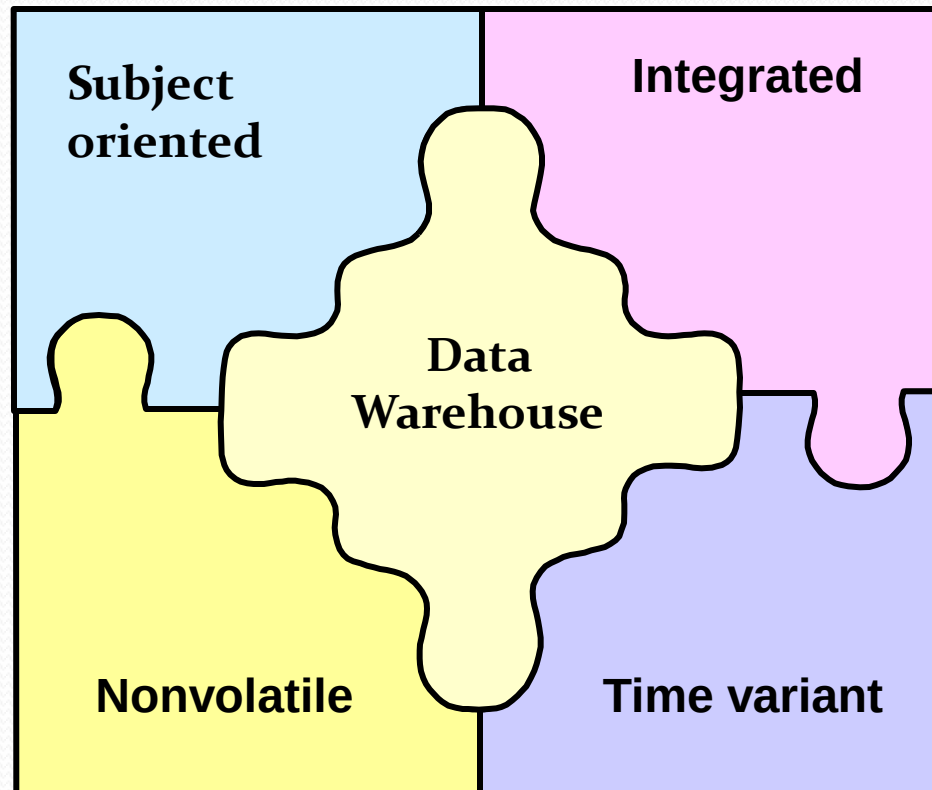
- BI is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies.
- BI a content-free expression, so it means different things to different people.
- BI helps *transform* data, to information (and knowledge), to decisions and finally to action.

# Data Warehouse: Definition

- A physical repository where relational data are specially organized to provide enterprise-wide, cleansed data in a standardized format
- “The data warehouse is a collection of integrated, subject-oriented databases designed to support DSS functions, where each unit of data is non-volatile and relevant to some moment in time”



# Data Warehouse Properties



# Subject Oriented

- Data is categorized and stored by business subject rather than by application.

## OLTP applications

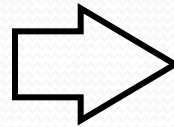
CDRs

Recharges

Invoices

Services

Loyalty points



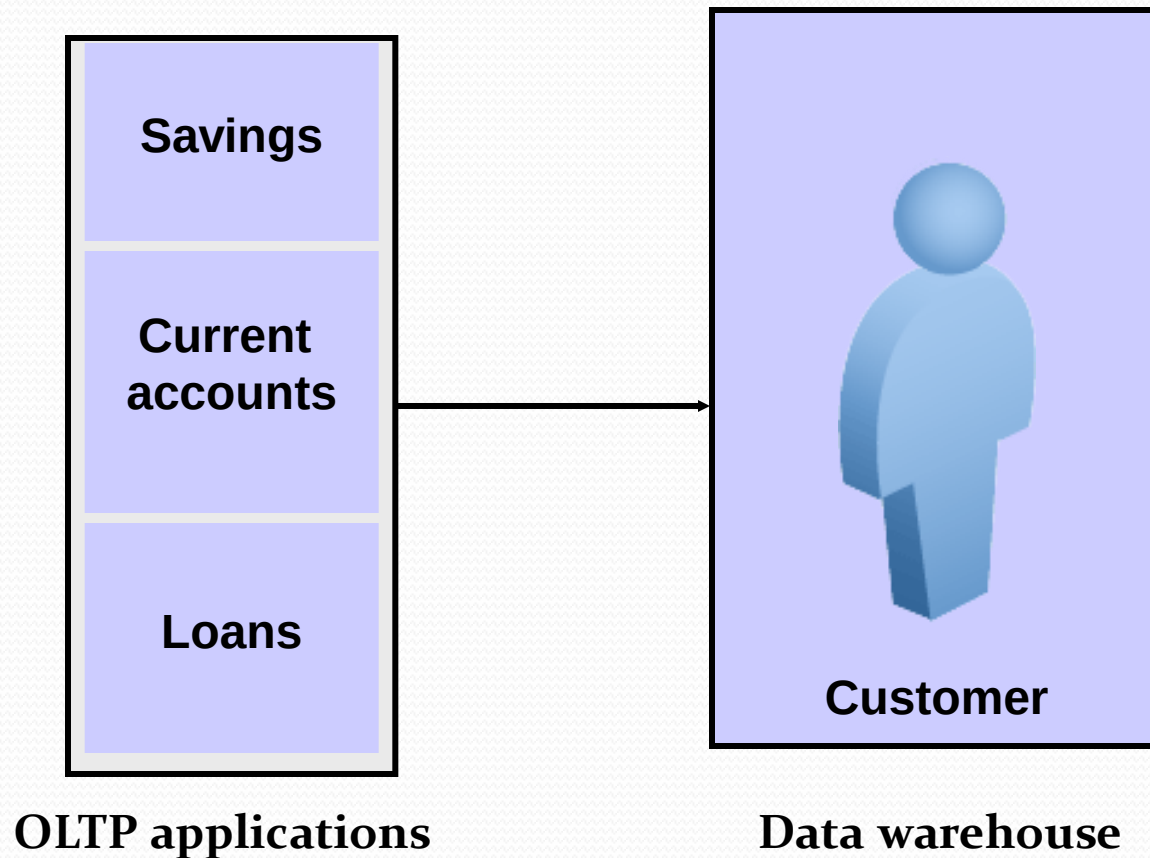
## Data warehouse subject



Customer Behavior  
information

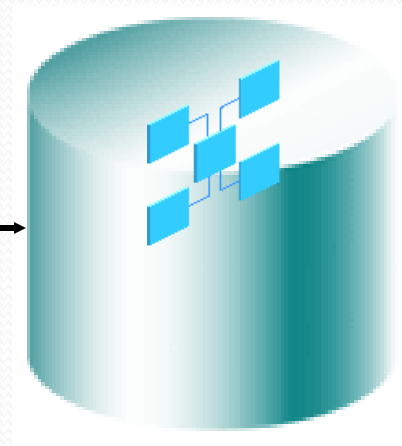
# Integrated

- Data on a given subject is defined and stored once.



# Time Variant

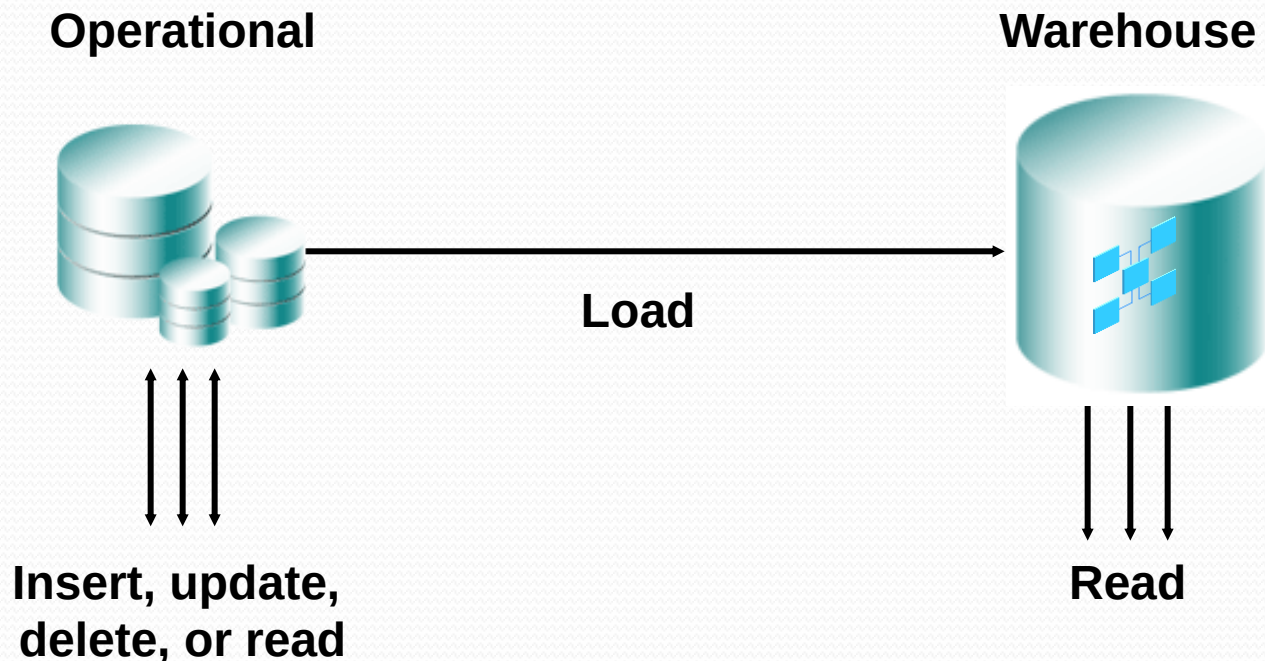
- Data is stored as a series of snapshots, each representing a period of time.



**Data  
warehouse**

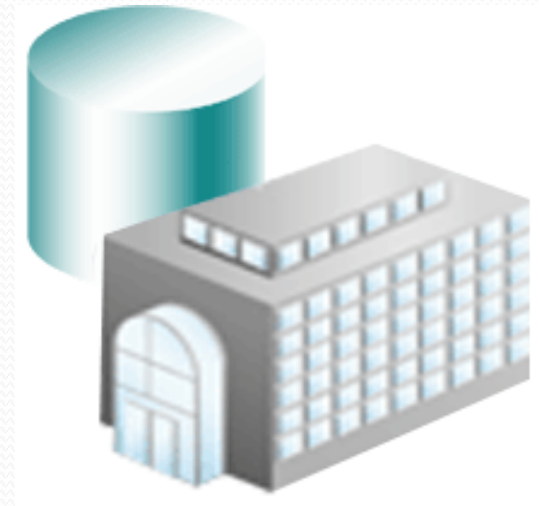
# Nonvolatile

- Typically, data in the data warehouse is not updated or deleted.



# Enterprise-Wide Data Warehouse

- Supports large-scale implementation
- Scopes the entire business
- Contains data from all subject areas
- Is developed incrementally
- Is a single source of enterprise-wide data
- Is the single distribution point to dependent data marts



# Data Mart

A departmental data warehouse that stores only relevant data

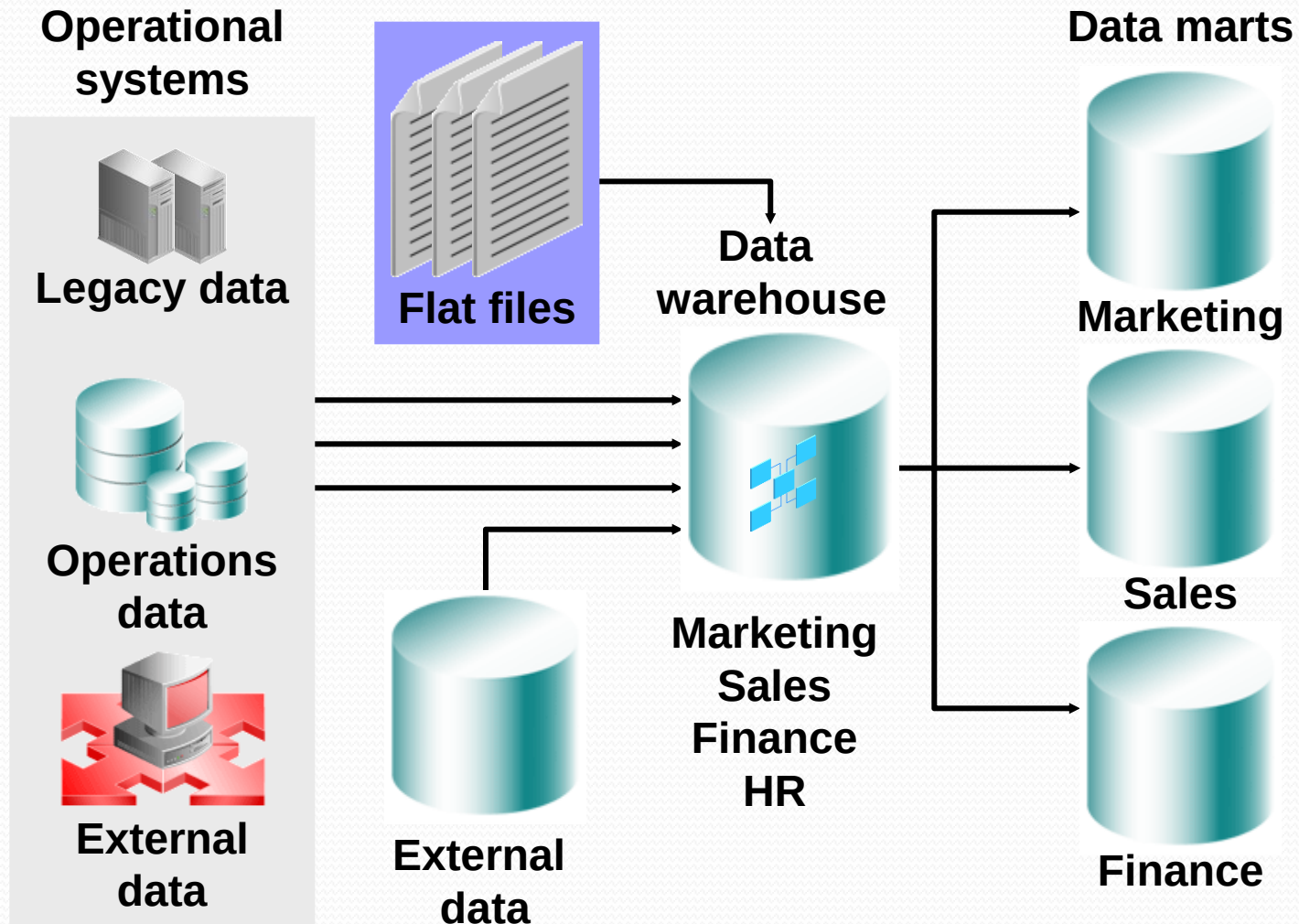
- **Dependent data mart**

A subset that is created directly from a data warehouse

- **Independent data mart**

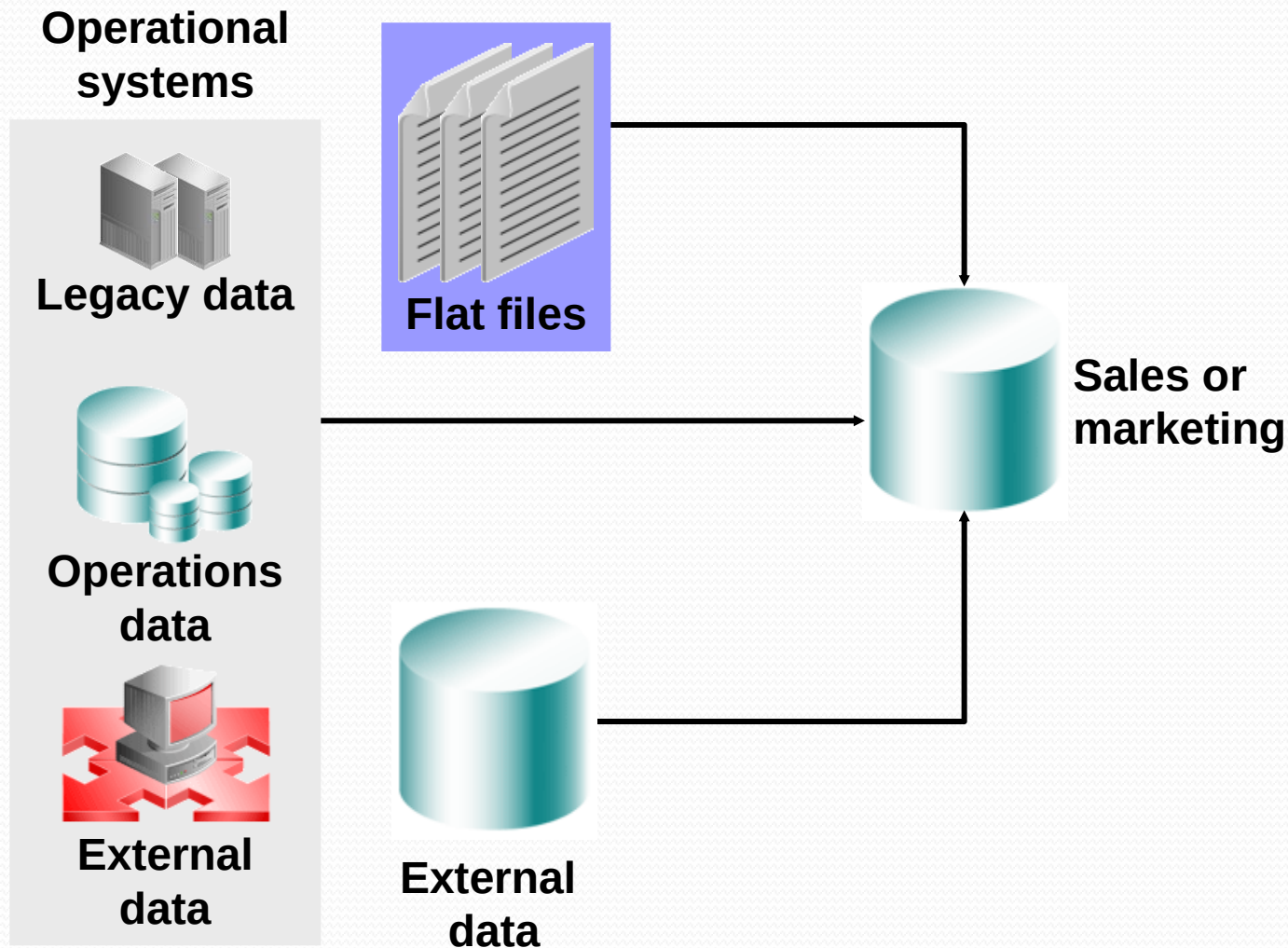
A small data warehouse designed for a strategic business unit or a department

# Dependent Data Mart





# Independent Data Mart



# Data Warehousing Definitions

- **Operational data stores (ODS)**

A type of database often used as an interim area for a data warehouse

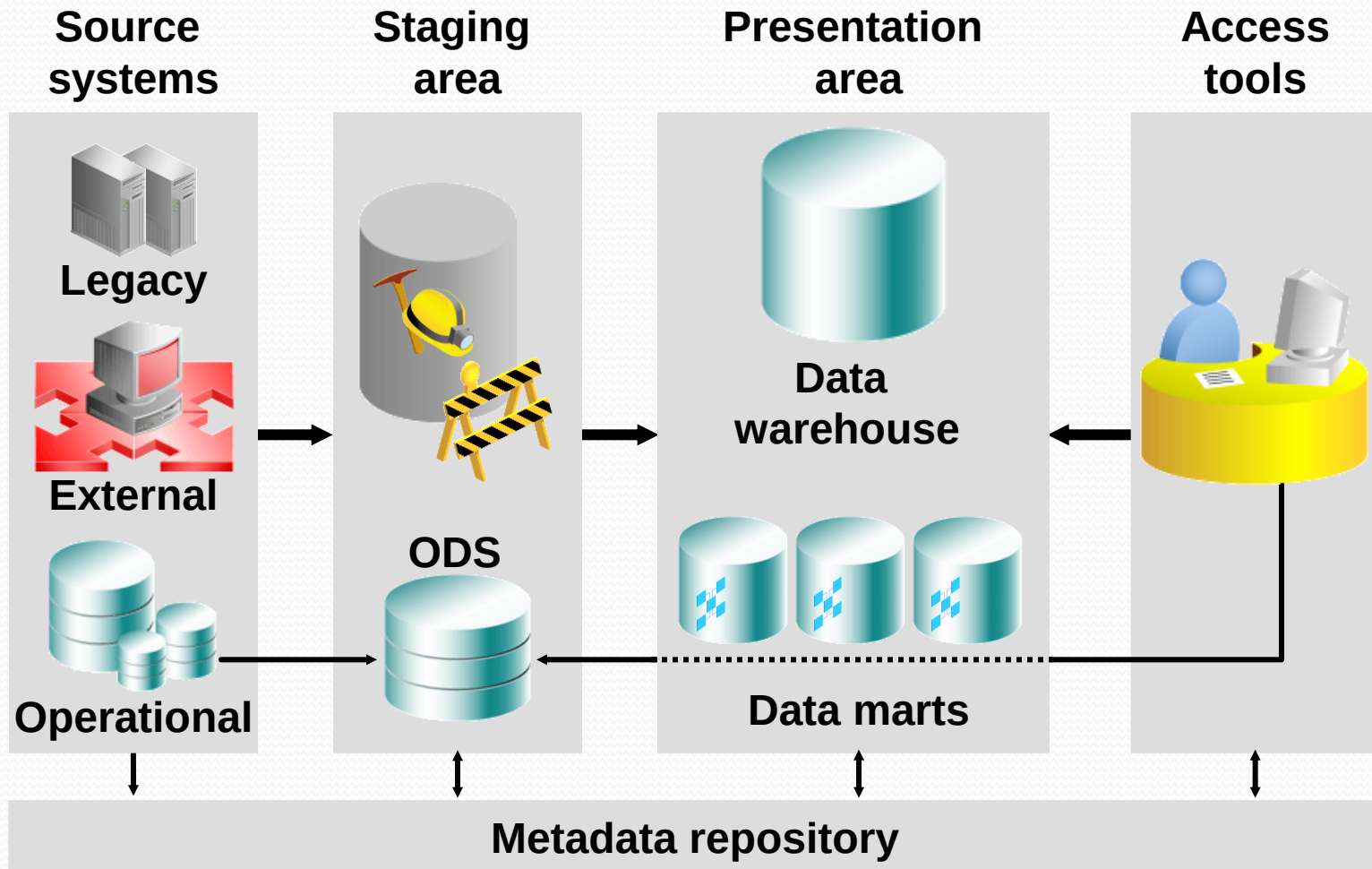
- **Enterprise data warehouse (EDW)**

A data warehouse for the enterprise

- **Metadata**

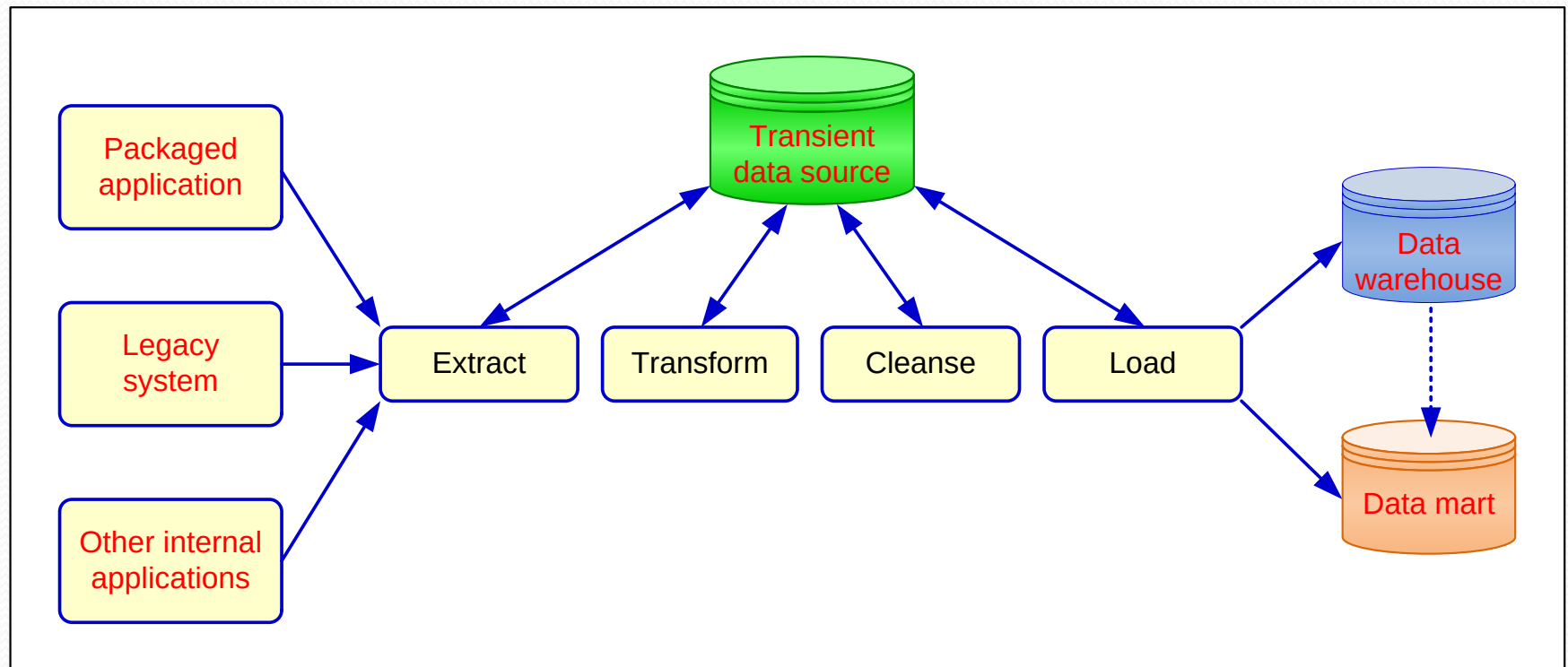
Data about data. In a data warehouse, metadata describe the contents of a data warehouse and the manner of its acquisition and use

# Typical Data Warehouse Components



# Data Integration and the Extraction, Transformation, and Load (ETL) Process

Extraction, transformation, and load (ETL)



# DW Architecture

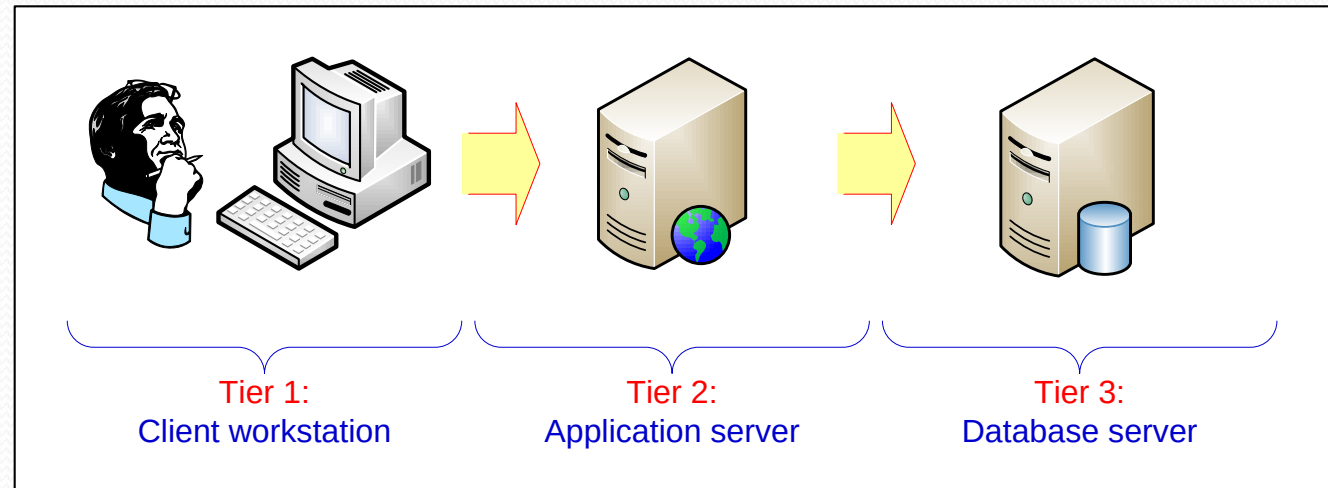
- **Three-tier architecture**
  1. Data acquisition software (back-end)
  2. The data warehouse that contains the data & software
  3. Client (front-end) software that allows users to access and analyze data from the warehouse
- **Two-tier architecture**

First 2 tiers in three-tier architecture is combined into one

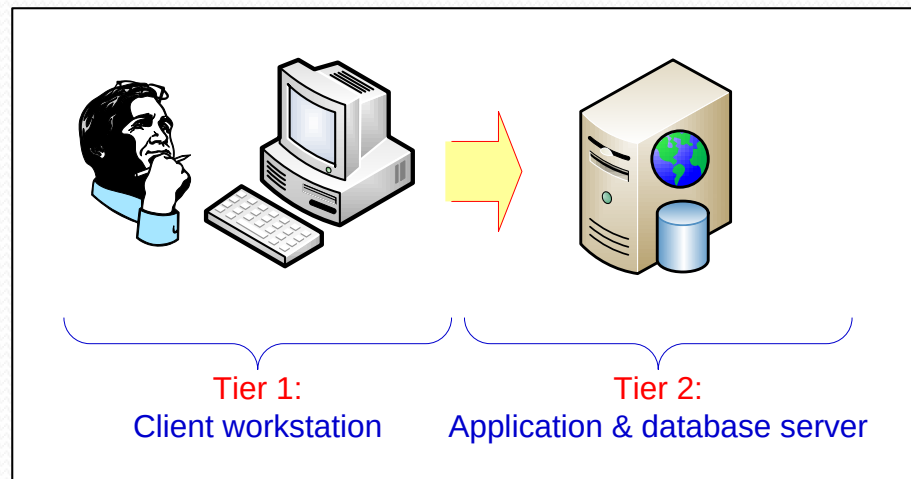
Sometimes there is only one tier

# DW Architectures

## 3-tier architecture

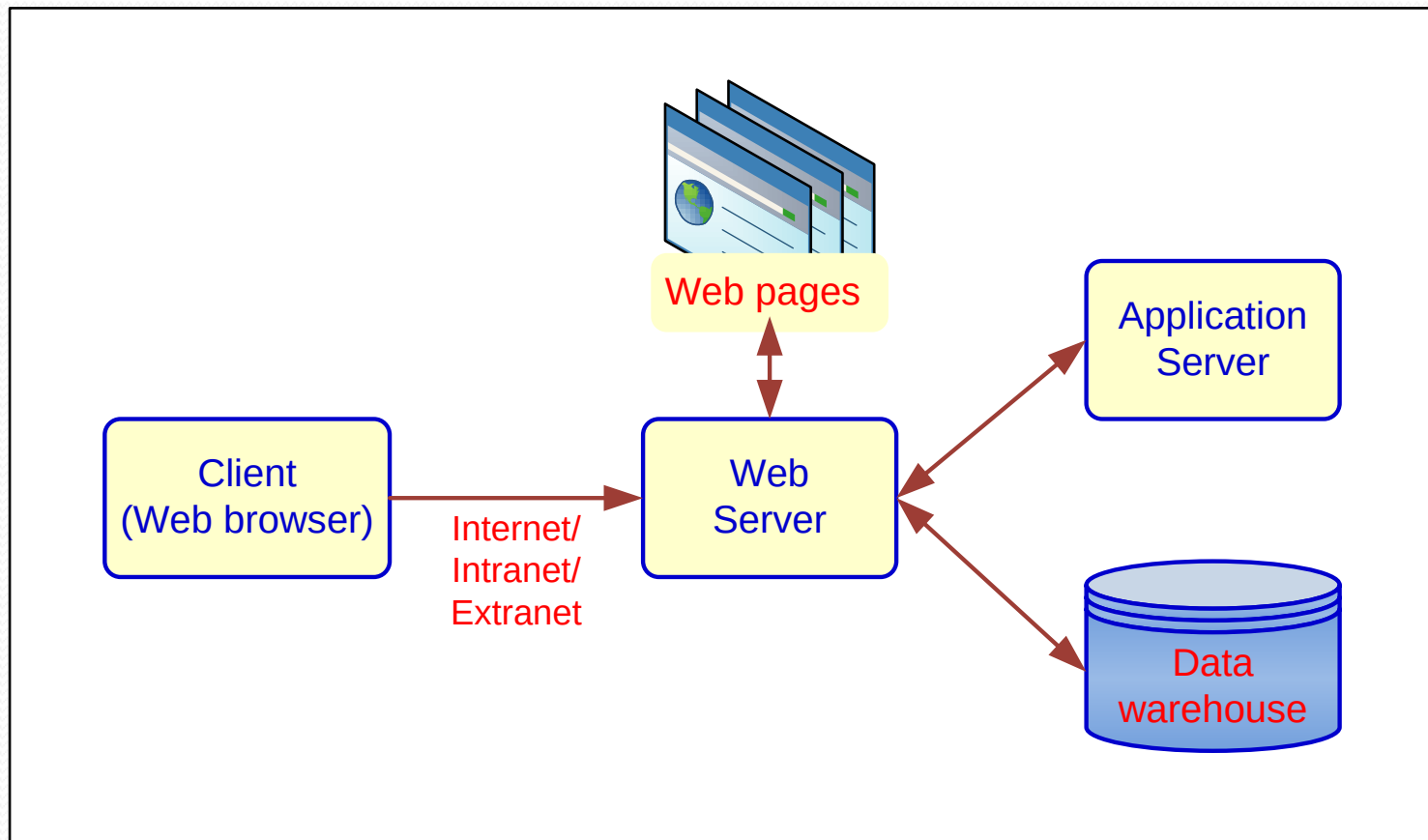


## 2-tier architecture



## 1-tier Architecture

# A Web-based DW Architecture



# Data Warehousing Architectures

- Issues to consider when deciding which architecture to use:
  - Which database management system (DBMS) should be used?
  - Will parallel processing and/or partitioning be used?
  - Will data migration tools be used to load the data warehouse?
  - What tools will be used to support data retrieval and analysis?

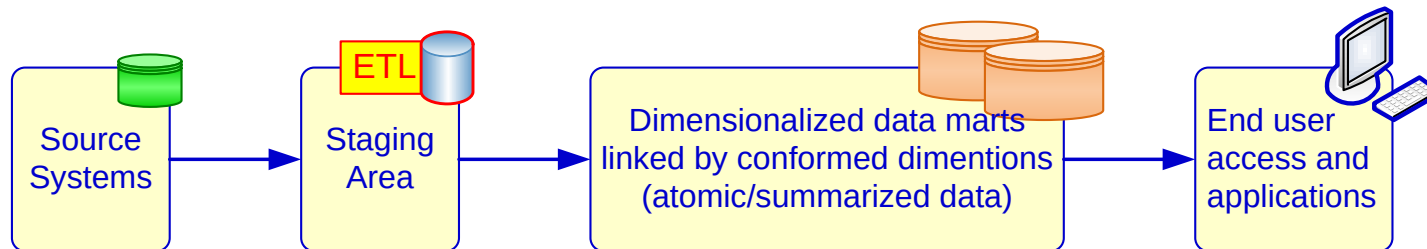


# Alternative DW Architectures

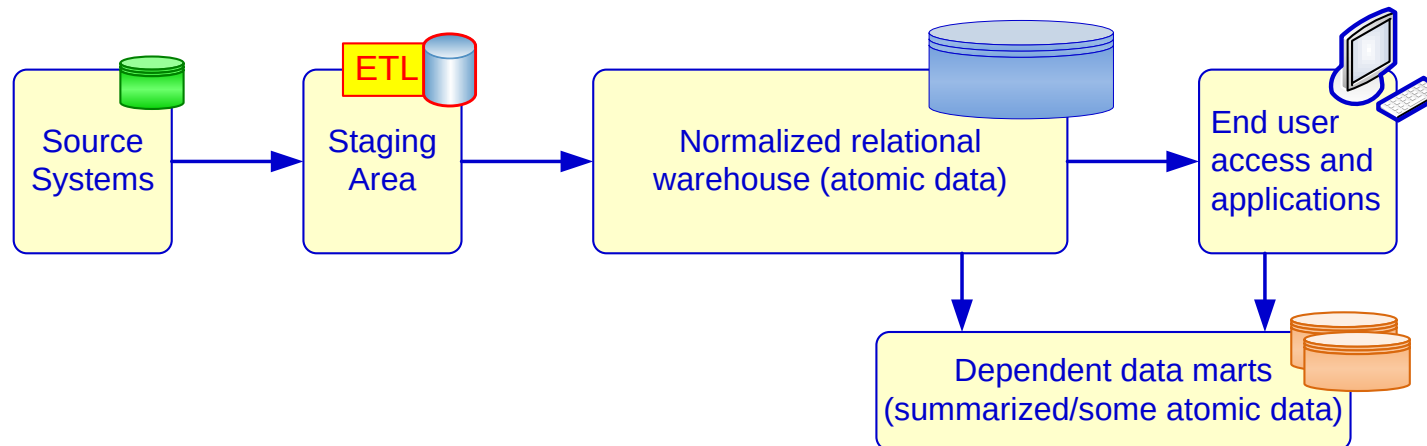
(a) Independent Data Marts Architecture



(b) Data Mart Bus Architecture with Linked Dimensional Datamarts

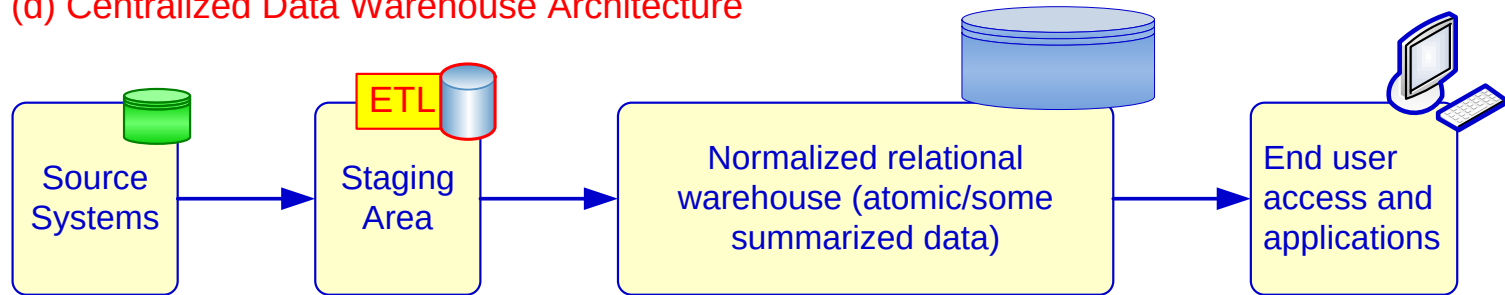


(c) Hub and Spoke Architecture (Corporate Information Factory)

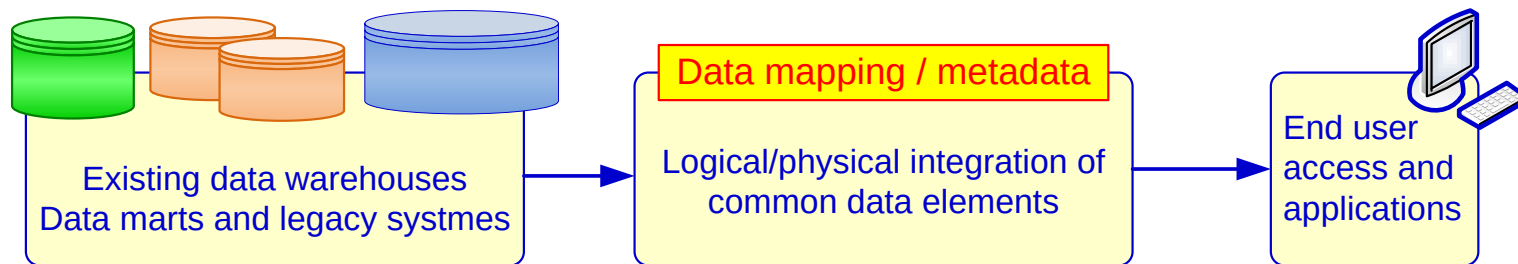


# Alternative DW Architectures

(d) Centralized Data Warehouse Architecture



(e) Federated Architecture



# DW Architectures

1. Independent Data Marts
  2. Data Mart Bus Architecture
  3. Hub-and-Spoke Architecture
  4. Centralized Data Warehouse
  5. Federated Data Warehouse
- Each has pros and cons!

# Data Warehousing Architectures

## Ten factors that potentially affect the architecture selection decision:

1. Information interdependence between organizational units
2. Upper management's information needs
3. Urgency of need for a data warehouse
4. Nature of end-user tasks
5. Constraints on resources
6. Strategic view of the data warehouse prior to implementation
7. Compatibility with existing systems
8. Perceived ability of the in-house IT staff
9. Technical issues
10. Social/political factors

# Data Warehouse Development

- Data warehouse development approaches
  - Inmon Model: EDW approach (top-down)
  - Kimball Model: Data mart approach (bottom-up)
  - Which model is best?
    - There is no one-size-fits-all strategy to DW
  - One alternative is the hosted warehouse
- Data warehouse structure:
  - The Star Schema vs. Relational
- Real-time data warehousing?

# Hosted Data Warehouses

- Benefits:
  - Requires minimal investment in infrastructure
  - Frees up capacity on in-house systems
  - Frees up cash flow
  - Makes powerful solutions affordable
  - Enables powerful solutions that provide for growth
  - Offers better quality equipment and software
  - Enables users to access data remotely
  - Allows a company to focus on core business
  - Meets storage needs for large volumes of data

# Representation of Data in DW

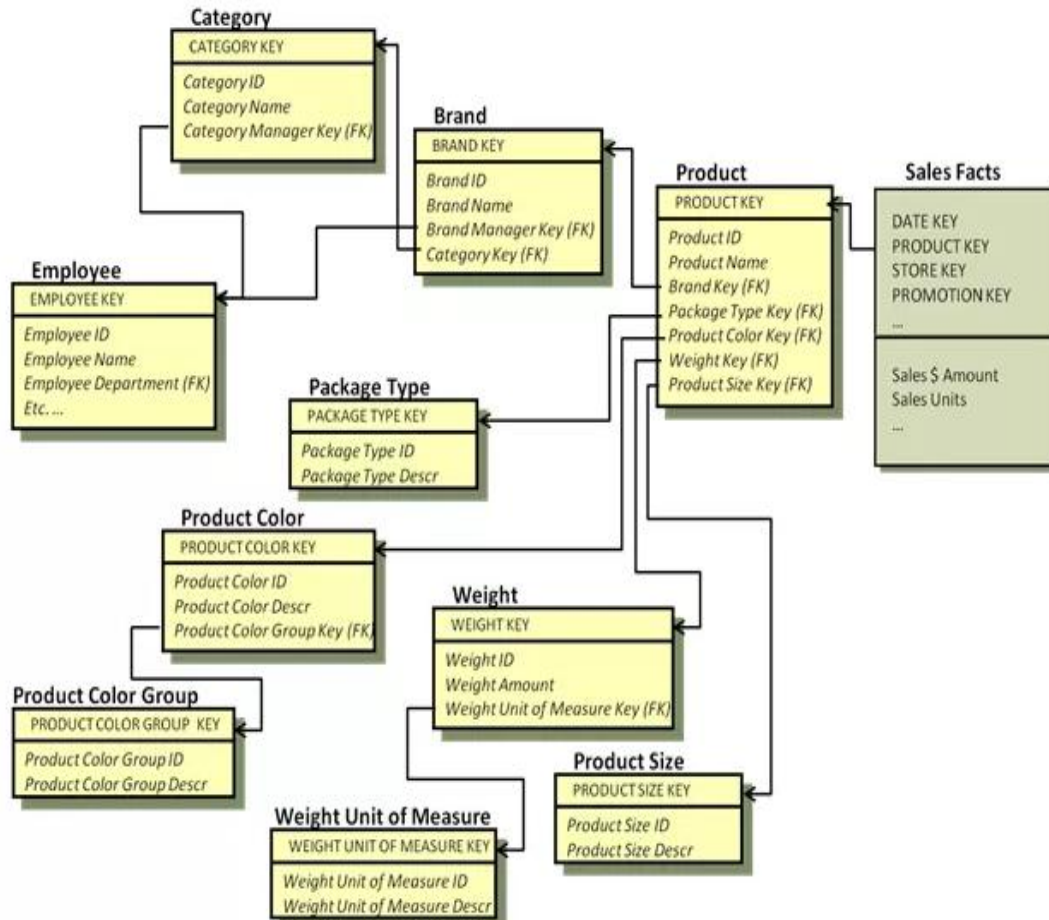
- **Dimensional Modeling** - the preferred technique for presenting analytic data because it addresses two simultaneous requirements:
  - Deliver data that's understandable to the business users.
  - Deliver fast query performance.
- A dimensional model contains the same information as a normalized model, but packages the data in a format that delivers user understandability, query performance, and resilience to change (the key difference is degree of normalization because both can be represented as ERDs)

# Analytical Business Questions ...

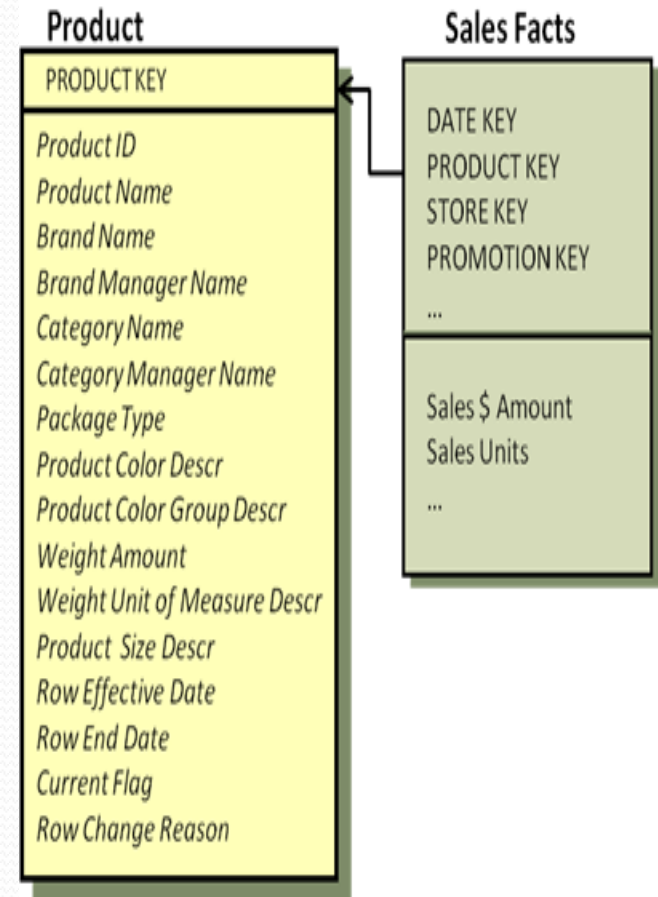
- How are sales year-to-date and how do they compare to last year?
- Who is most likely to respond to my current marketing campaign and how will they impact revenue?
- What is the turnover in employees compared to the last five years?
- How is potential fraud cost being managed over time?
- What are my most profitable products by region, by year, and year-to-date?



# Relational Vs Dimensional Models



Relational Model



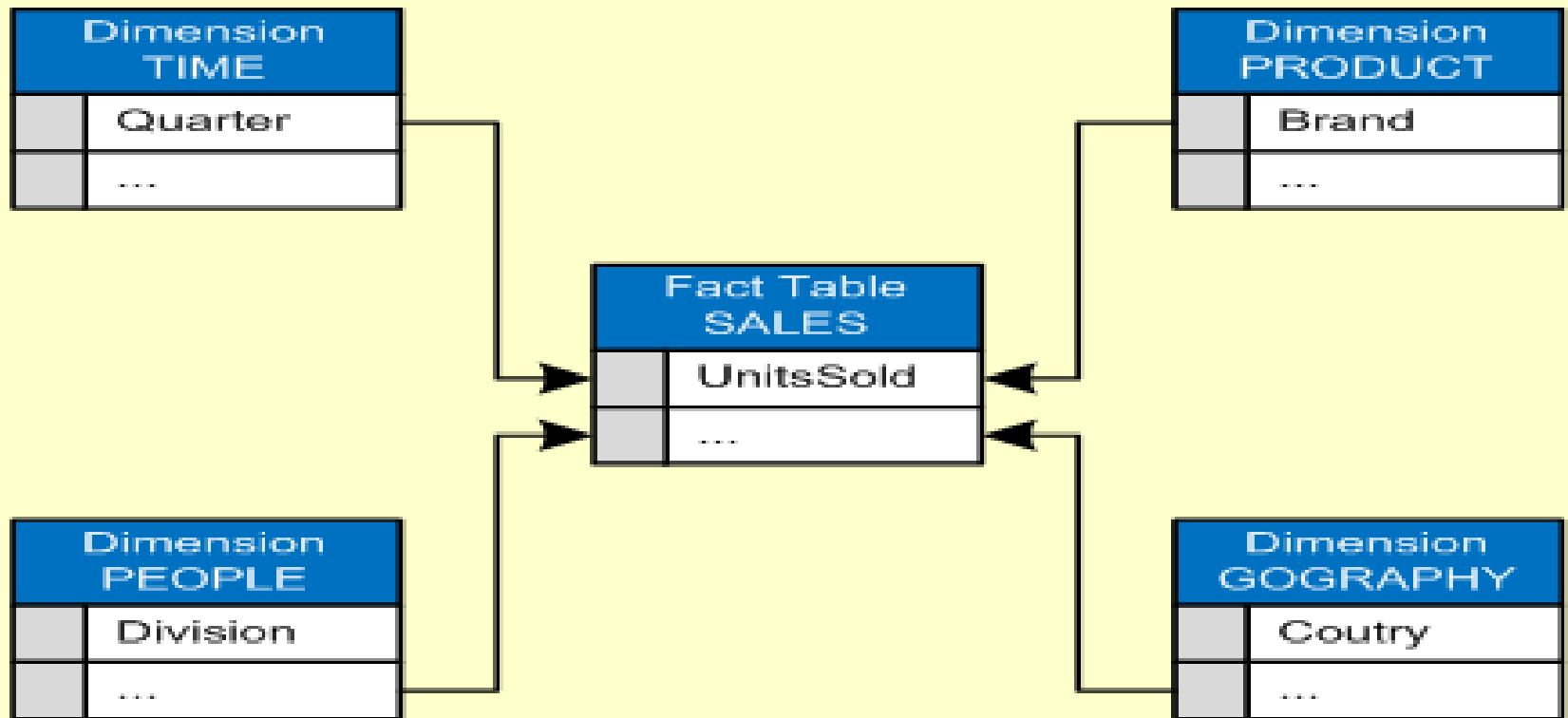
Dimensional Model

# Representation of Data in DW

- **Dimensional Modeling** – a retrieval-based system that supports high-volume query access
- **Star schema** – the most commonly used and the simplest style of dimensional modeling
  - Contain a **fact table** surrounded by and connected to several **dimension tables**
  - Fact table contains the descriptive attributes (numerical values) needed to perform decision analysis and query reporting
  - Dimension tables contain classification and aggregation information about the values in the fact table
- **Snowflakes schema** – an extension of star schema where the diagram resembles a snowflake in shape

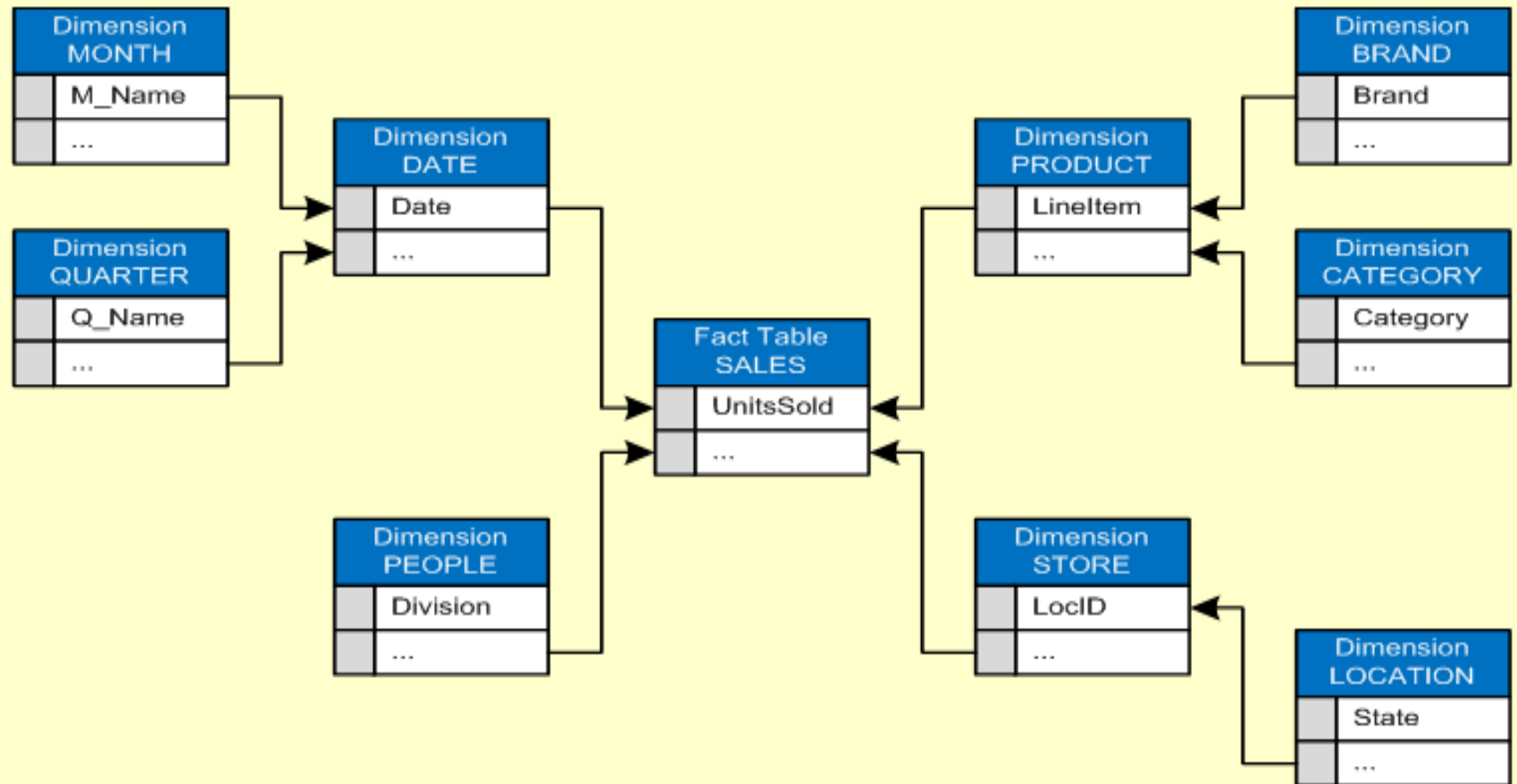
# Star Schema

## Star Schema



# Snowflake Schema

## Snowflake Schema



# Multidimensionality

- Multidimensionality

The ability to organize, present, and analyze data by several dimensions, such as sales by *region*, by *product*, by *salesperson*, and by *time* (four dimensions)

- Multidimensional presentation

- **Dimensions:** products, salespeople, market segments, business units, geographical locations, distribution channels, country, or industry
- **Measures:** money, sales volume, head count, inventory profit, actual versus forecast

# Analysis of Data in DW

- Online analytical processing (OLAP)
  - Data driven activities performed by end users to query the online system and to conduct analyses
  - Data cubes, drill-down / rollup, slice & dice, ...
- OLAP Activities
  - Generating queries (query tools)
  - Requesting ad hoc reports
  - Conducting statistical and other analyses
  - Developing multimedia-based applications

# Analysis of Data Stored in DW

## OLTP vs. OLAP

- OLTP (online transaction processing)
  - A system that is primarily responsible for capturing and storing data related to day-to-day business functions such as ERP, CRM, SCM, POS,
  - The main focus is on efficiency of routine tasks
- OLAP (online analytic processing)
  - A system is designed to address the need of information extraction by providing effectively and efficiently ad hoc analysis of organizational data
  - The main focus is on effectiveness

# OLAP Versus OLTP

| Property          | OLTP                  | OLAP                            |
|-------------------|-----------------------|---------------------------------|
| Response time     | Subseconds to seconds | Seconds to hours                |
| Operations        | DML                   | Primarily read-only             |
| Nature of data    | 30–60 days            | Snapshots over time             |
| Data organization | Application           | Subject, time                   |
| Size              | Small to large        | Large to very large             |
| Data sources      | Operational, internal | Operational, internal, external |
| Activities        | Processes             | Analysis                        |



# Variations of OLAP

- **Multidimensional OLAP (MOLAP)**

OLAP implemented via a specialized multidimensional database (or data store) that summarizes transactions into multidimensional views ahead of time

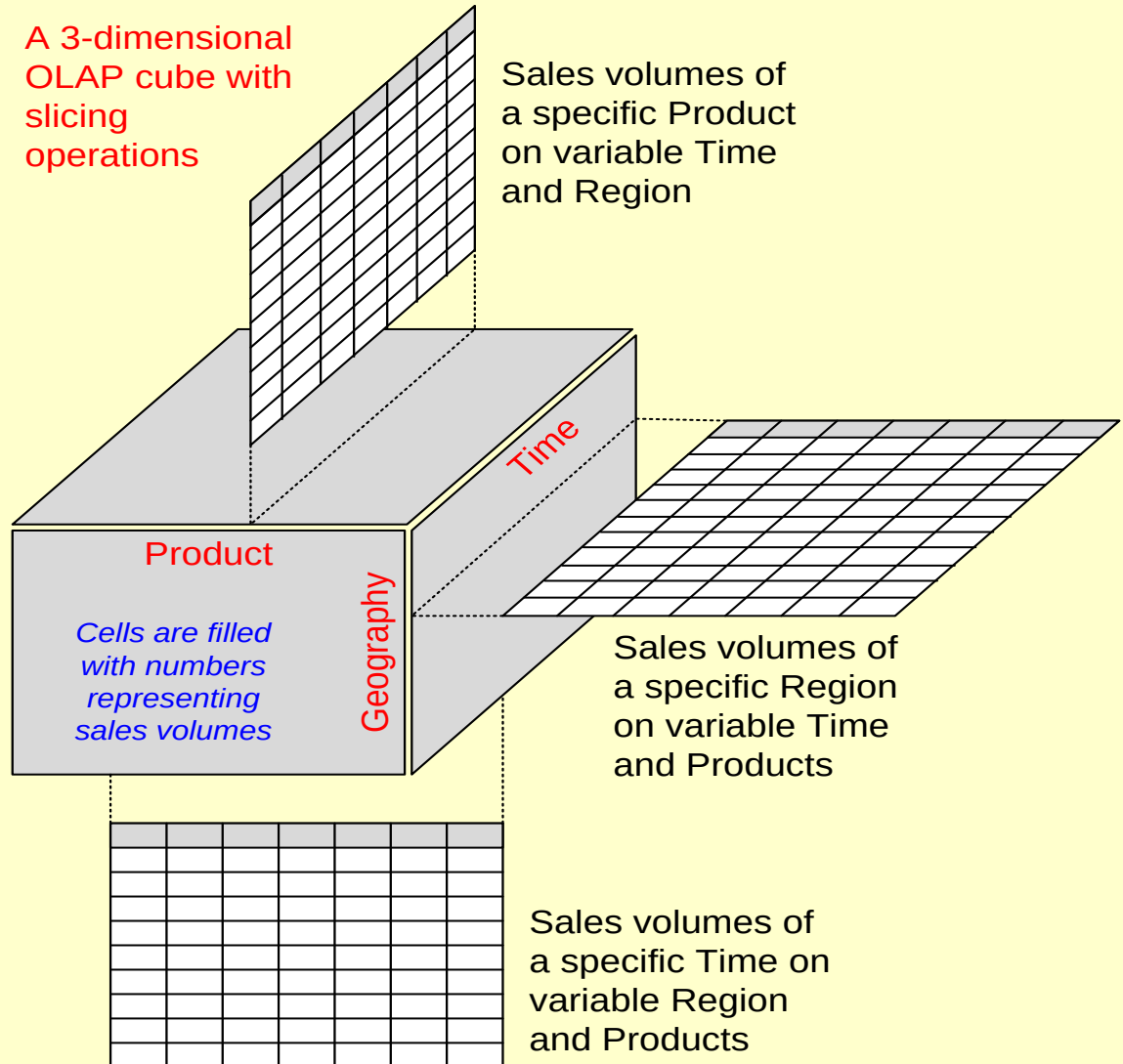
- **Relational OLAP (ROLAP)**

The implementation of an OLAP database on top of an existing relational database

# OLAP

## Slicing Operations on a Simple Tree-Dimensional Data Cube

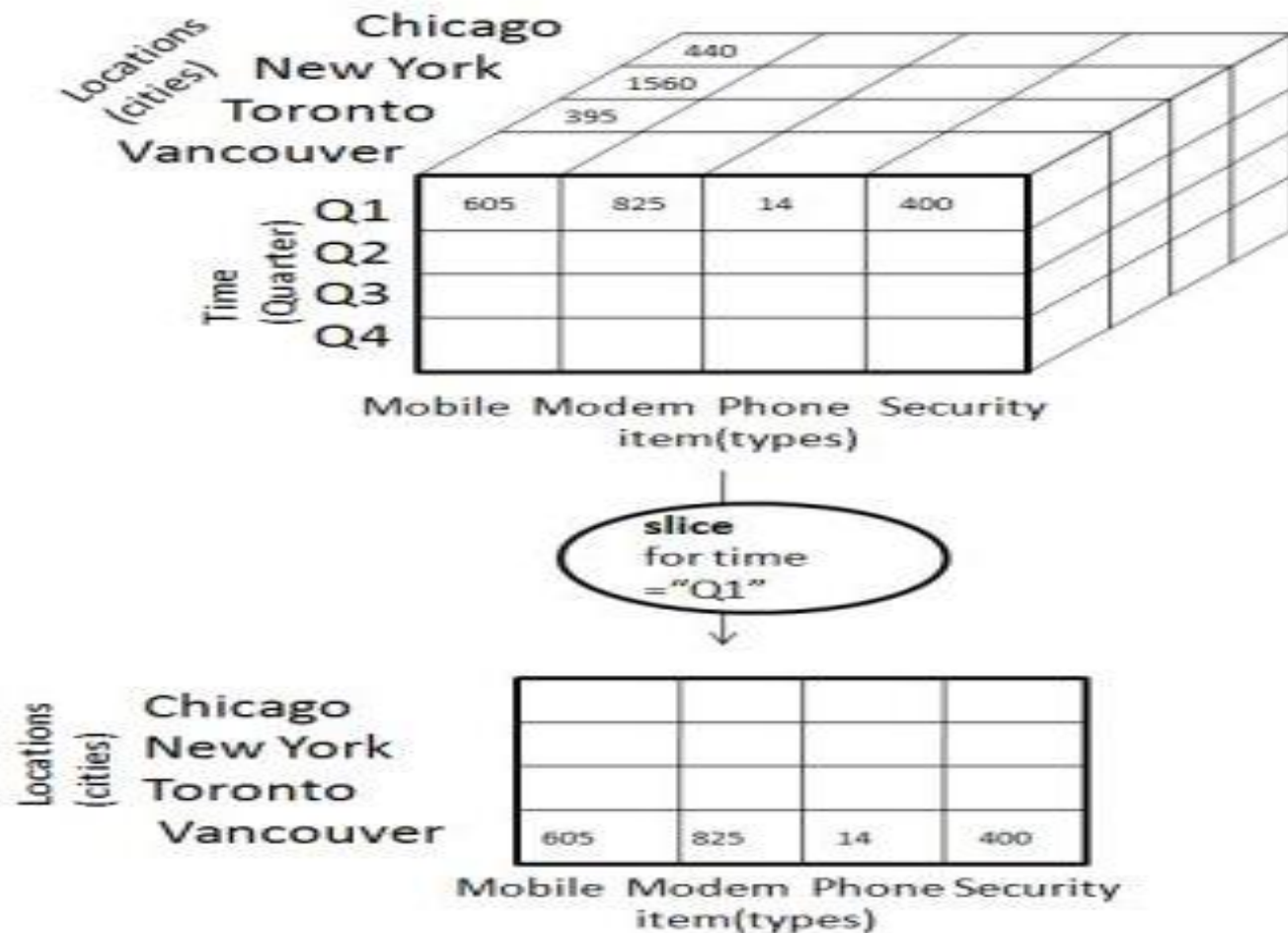
A 3-dimensional  
OLAP cube with  
slicing  
operations



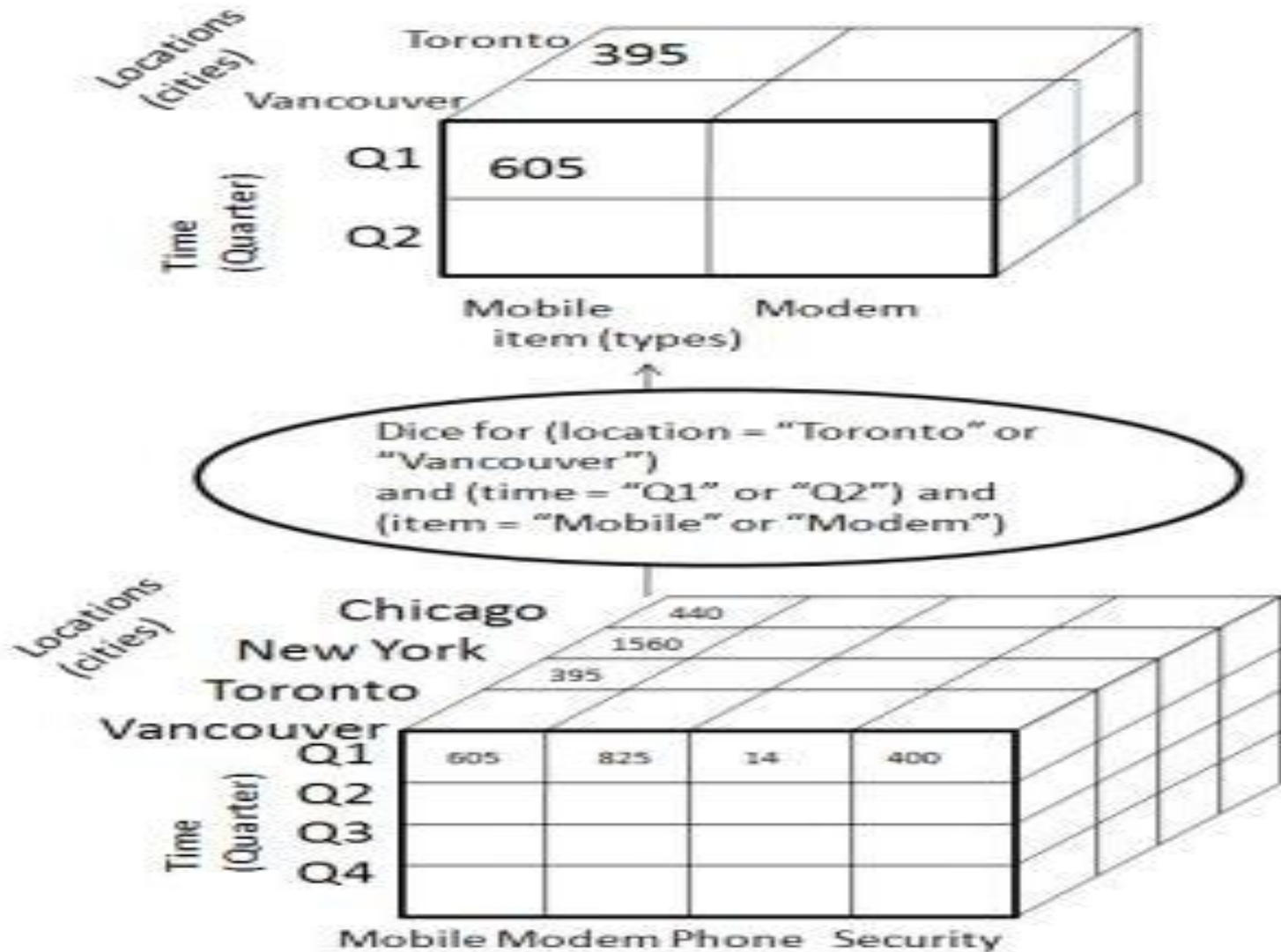
# OLAP Operations

- **Slice** – a subset of a multidimensional array
- **Dice** – a slice on more than two dimensions
- **Drill Down** – navigating among levels of data ranging from the most summarized (up) to the most detailed (down)
- **Roll Up** – computing all of the data relationships for one or more dimensions
- **Pivot** – used to change the dimensional orientation of a report or an ad hoc query-page display

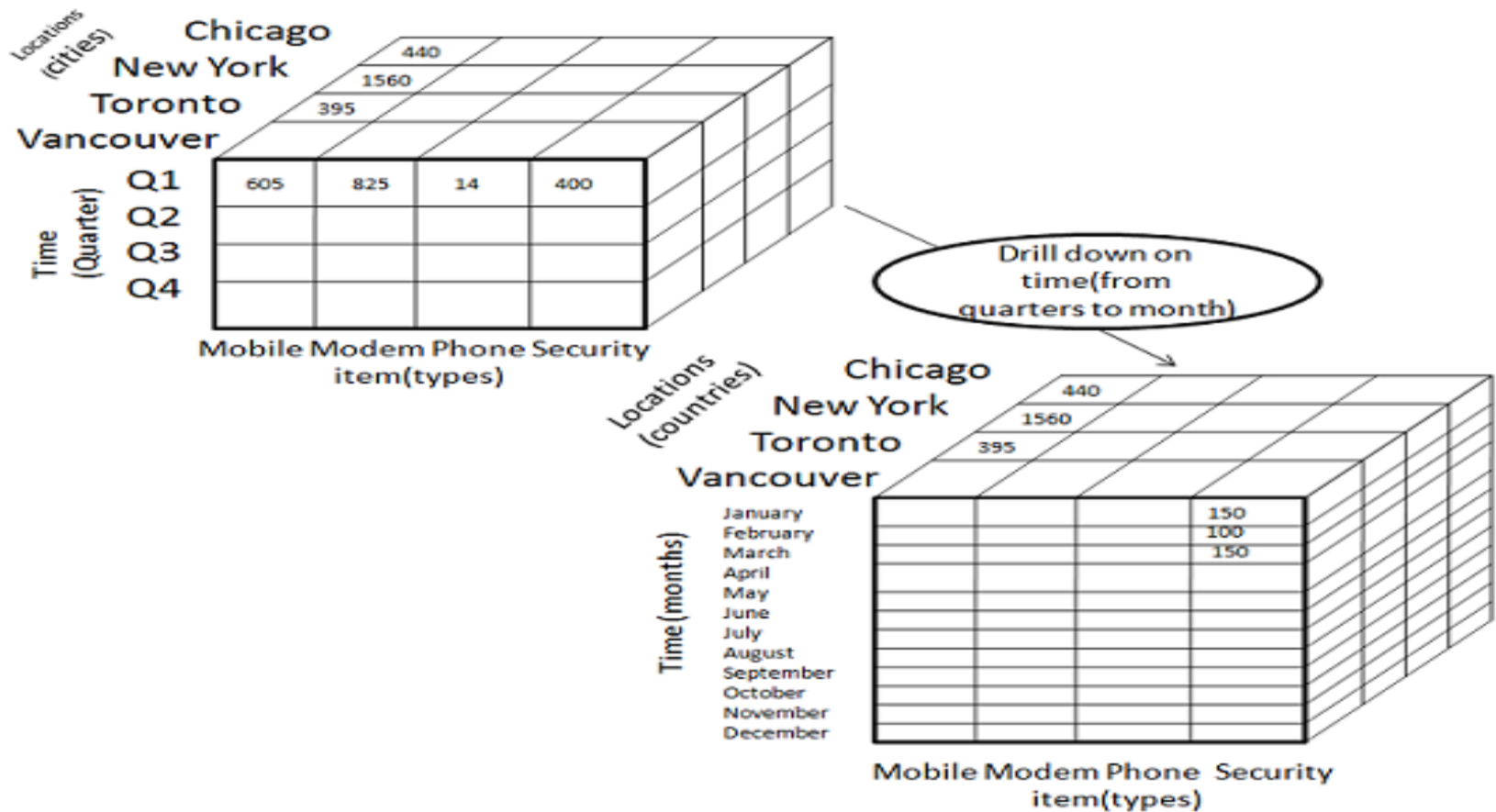
# Slice



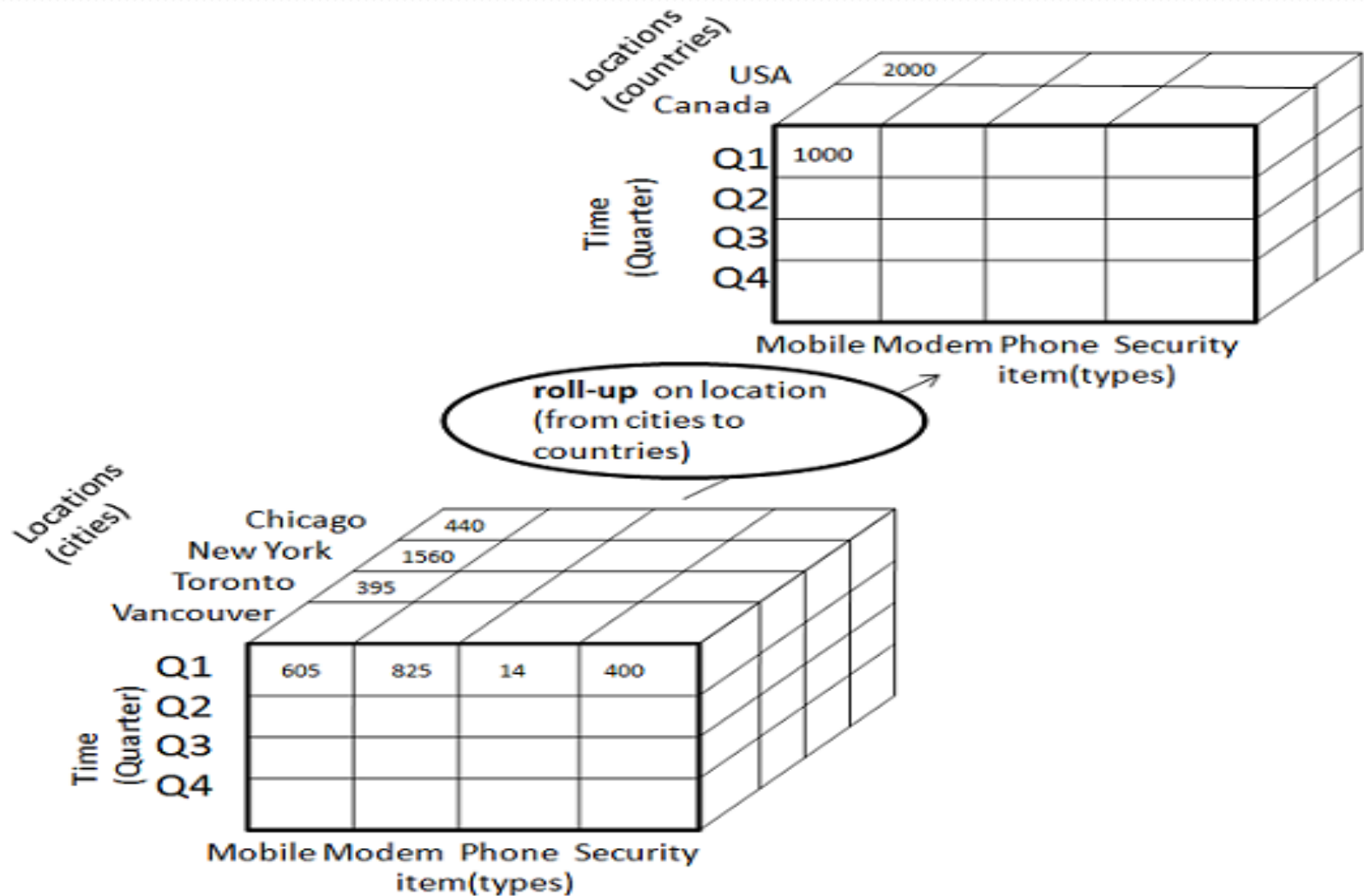
# Dice



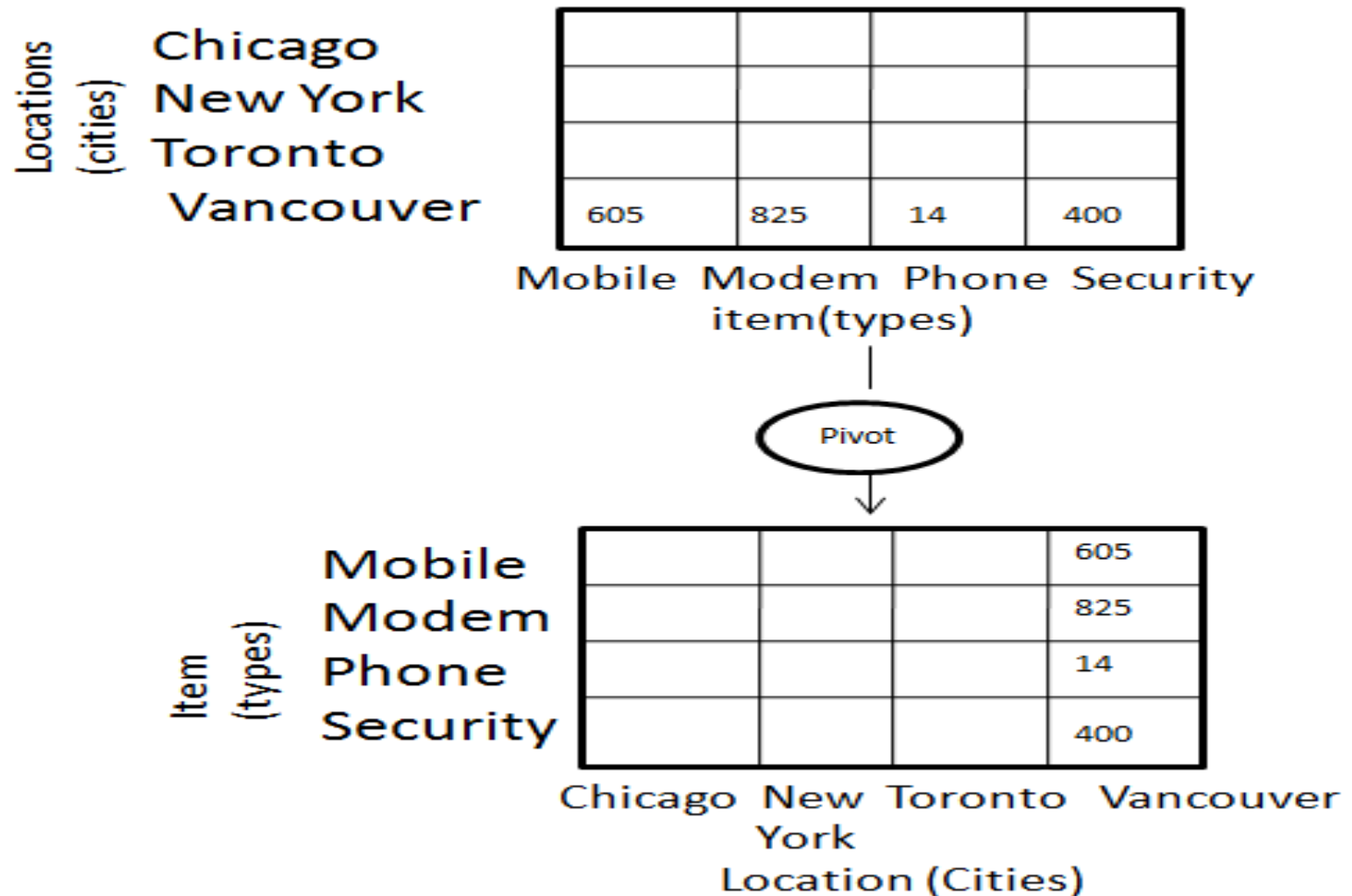
# Drill Down



# Roll up



# Pivot







# Chapter 3

## Dimensional Modeling (Retail Sales)

# Dimensional Modeling Process

## Step 1: Select the Business Process

- A *business process* is a low-level activity performed by an organization, such as **taking orders, invoicing, receiving payments, handling service calls, registering students** or **performing a medical procedure**.

## Step 2: Declare the Grain

- specifying exactly what an individual fact table row represents. The grain conveys the level of detail associated with the fact table measurements.

## Step 3: Identify the Dimensions

- Identify all possible descriptions that take on single values in the context of each measurement.

## Step 4: Identify the Facts

- Identify all performance metrics to be measures, such as sales revenue, sales quantity, Gross Profit

# Retail Case Study

- The business has 100 grocery stores spread across five states. Each store has a full complement of departments, including grocery, frozen foods, dairy, meat, produce, bakery, floral and health/beauty aids.
- Each store has approximately 60,000 individual products, called *stock keeping units* (*SKUs*), on its shelves.
- Data is collected from *point-of-sales* (POS) system.

# Retail Case Study

Allstar Grocery  
123 Loon Street  
Green Prairie, MN 55555  
(952) 555-1212

Store: 0022  
Cashier: 00245409/Alan

|                                          |       |
|------------------------------------------|-------|
| 0030503347 Baked Well Multigrain Muffins | 2.50  |
| 2120201195 Diet Cola 12-pack             | 4.99  |
| Saved \$.50 off \$5.49                   |       |
| 0070806048 Sparkly Toothpaste            | 1.99  |
| Coupon \$.30 off \$2.29                  |       |
| 2840201912 SoySoy Milk Quart             | 3.19  |
| TOTAL                                    | 12.67 |
| AMOUNT TENDERED                          |       |
| CASH                                     | 12.67 |
| ITEM COUNT:                              | 4     |

-----  
Transaction: 649

4/15/2013 10:56 AM  
-----

Thank you for shopping at Allstar  
0064900220415201300245409

# Retail Case Study (Cont.)

- Step 1: Select the Business Process
  - The first DW/BI project should focus on the business process that is both the most critical to the business users, as well as the most feasible. Feasibility covers a range of considerations, including data availability and quality, as well as organizational readiness.
  - The business process you're modeling is POS retail sales transactions. This data enables the business users to analyze which products are selling in which stores on which days under what promotional conditions in which transactions.

# Retail Case Study (Cont.)

- Step 2: Declare the Grain
  - Identify **level of data detail** should be made available in the dimensional model (daily, weekly, monthly, etc.)
  - **Detailed/Atomic** data in dimensional model is poised and ready for the **ad hoc attack** by business users. It provides the maximum analytic flexibility.
  - **Aggregated /summarized** data plays an important role in **performance tuning**
  - Retail transaction atomic data is up to time stamp level

# Retail Case Study (Cont.)

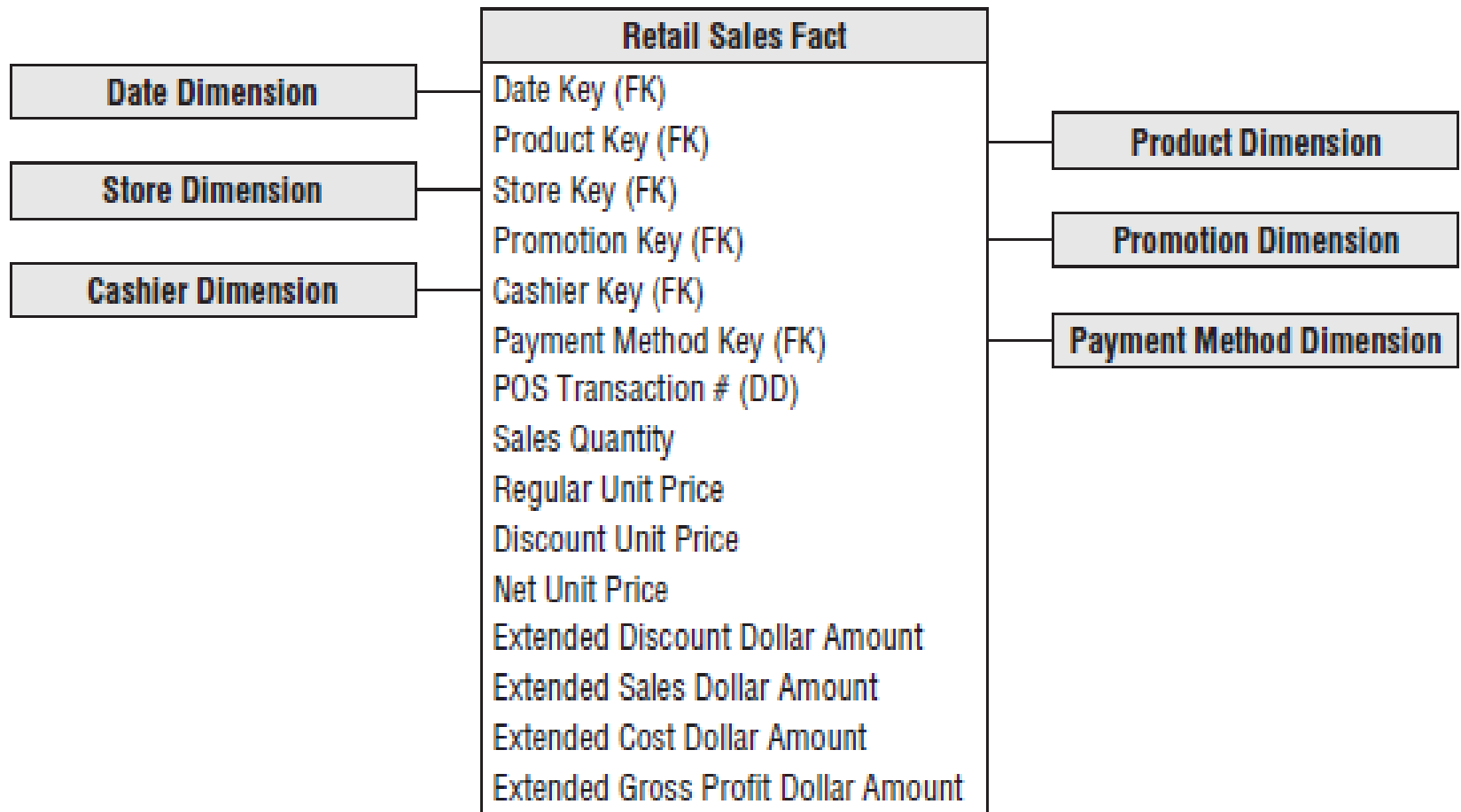
- Step 3: Identify the Dimensions
  - A careful **grain** statement **determines the primary dimensionality** of the fact table. You then add more dimensions to the fact table if these additional dimensions naturally take on only one value under each combination of the primary dimensions. If the additional dimension violates the grain by causing additional fact rows to be generated, the dimension needs to be disqualified or the grain statement needs to be revisited.
  - Possible dimensions are date, product, store, promotion, cashier, and method of payment, POS transaction ticket number

# Retail Case Study (Cont.)

- Step 4: Identify the Facts
  - The facts indicates the measures or KPIs (Key Performance Indicators) that shall be measured.
  - Possible Facts collected by POS system are sales quantity (for example, the number of cans of chicken noodle soup), discount, and net paid prices, extended discounts and sales dollar amounts (Total sales = sales quantity \* sold amount per item)



# Retail Sales Model (Star Schema)



# Fact Types

1. **Derived Facts** (Ex: Extended Sales, Extended Gross Profit)
  - calculate its value based on other stored attributes/facts
  - It's recommended to be **stored** physically to better **performance** in querying as well as **calculation consistency**
  - Some organizations want to perform the calculation in the **BI tool**
2. **Non-Additive Facts** (Ex: Gross Margin = Gross Profit/Extended Sales)
  - **Percentages and ratios**, such as gross margin, are non-additive. The numerator and denominator should be stored in the fact table
  - These non-additive facts may be averaged carefully over many records, if the business analysts agree that this makes sense.
3. **Additive Facts** (Ex: Sales Amount, Sales Quantity)

# Dimension Table Details

## (Date Dimension)

- The date dimension is a special dimension because it is the one dimension nearly guaranteed to be in every dimensional model since virtually every business process **captures a time series of performance metrics**.
- Dimensional models always need an **explicit** date **dimension** table. There are many date attributes not supported by the SQL date function, including week numbers, fiscal periods, seasons, holidays, and weekends.
- **Time** is included in a Dimension or Fact ?

| Date Dimension                |
|-------------------------------|
| Date Key (PK)                 |
| Date                          |
| Full Date Description         |
| Day of Week                   |
| Day Number in Calendar Month  |
| Day Number in Calendar Year   |
| Day Number in Fiscal Month    |
| Day Number in Fiscal Year     |
| Last Day in Month Indicator   |
| Calendar Week Ending Date     |
| Calendar Week Number in Year  |
| Calendar Month Name           |
| Calendar Month Number in Year |
| Calendar Year-Month (YYYY-MM) |
| Calendar Quarter              |
| Calendar Year-Quarter         |
| Calendar Year                 |
| Fiscal Week                   |
| Fiscal Week Number in Year    |
| Fiscal Month                  |
| Fiscal Month Number in Year   |
| Fiscal Year-Month             |
| Fiscal Quarter                |
| Fiscal Year-Quarter           |
| Fiscal Half Year              |
| Fiscal Year                   |
| Holiday Indicator             |
| Weekday Indicator             |

# Dimension Table Details

## (Product Dimension)

- Natural Keys
  - Operational product codes have embedded meaning with different parts of the code representing significant characteristics of the product (Ex: sku ...)
- Surrogate Keys
  - Unique primary key of a dimension table belongs to DW models only
  - Join dimension tables to fact tables
  - Faster performance, Avoid operational changes and integration of multiple sources
  - Can be used for fact tables as a primary key

| Product Dimension        |
|--------------------------|
| Product Key (PK)         |
| SKU Number (NK)          |
| Product Description      |
| Brand Description        |
| Subcategory Description  |
| Category Description     |
| Department Number        |
| Department Description   |
| Package Type Description |
| Package Size             |
| Fat Content              |
| Diet Type                |
| Weight                   |
| Weight Unit of Measure   |
| Storage Type             |
| Shelf Life Type          |
| Shelf Width              |
| Shelf Height             |
| Shelf Depth              |
| ...                      |

# Dimension Table Details

## (Store Dimension)

- Multiple Hierarchies in Dimension Tables
  - The store dimension is the case study's primary **geographic dimension**. Each store can be thought of as a location
  - It is not uncommon to represent multiple hierarchies in a dimension table. The attribute names and values should be unique across the multiple hierarchies.
- Dates Within Dimension Tables
  - The first open date and last remodel date in the store dimension could be date type columns. However, if users want to group and constrain on nonstandard calendar attributes (like the open date's fiscal period), then they are typically join keys to copies of the date dimension table.

| Store Dimension        |
|------------------------|
| Store Key (PK)         |
| Store Number (NK)      |
| Store Name             |
| Store Street Address   |
| Store City             |
| Store County           |
| Store City-State       |
| Store State            |
| Store Zip Code         |
| Store Manager          |
| Store District         |
| Store Region           |
| Floor Plan Type        |
| Photo Processing Type  |
| Financial Service Type |
| Selling Square Footage |
| Total Square Footage   |
| First Open Date        |
| Last Remodel Date      |
| ...                    |

# More Dimensions ....

- Promotion Dimension (Causal dimension)
- Sales Agent Dimension
- Payment Method Dimension
- Customer Dimension

# Retail Schema in action

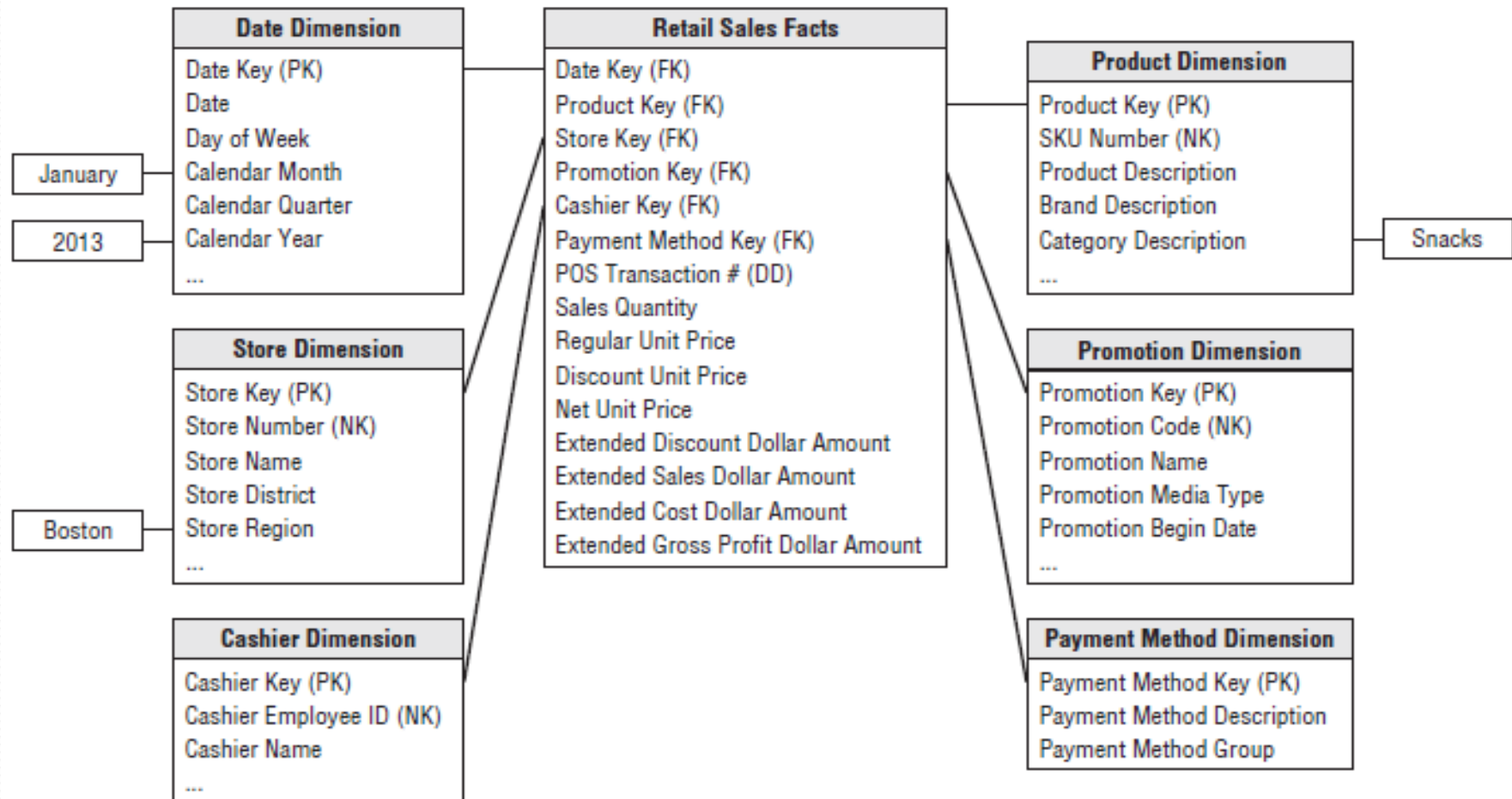


| <b>Calendar Week<br/>Ending Date</b> | <b>Promotion Name</b> | <b>Extended Sales<br/>Dollar Amount</b> |
|--------------------------------------|-----------------------|-----------------------------------------|
| January 6, 2013                      | No Promotion          | 2,647                                   |
| January 13, 2013                     | No Promotion          | 4,851                                   |
| January 20, 2013                     | Super Bowl Promotion  | 7,248                                   |
| January 27, 2013                     | Super Bowl Promotion  | 13,798                                  |

| <b>Department<br/>Name</b> | <b>No Promotion<br/>Extended Sales<br/>Dollar Amount</b> | <b>Super Bowl Promotion<br/>Extended Sales<br/>Dollar Amount</b> |
|----------------------------|----------------------------------------------------------|------------------------------------------------------------------|
| January 6, 2013            | 2,647                                                    | 0                                                                |
| January 13, 2013           | 4,851                                                    | 0                                                                |
| January 20, 2013           | 0                                                        | 7,248                                                            |
| January 27, 2013           | 0                                                        | 13,798                                                           |



# Retail Schema in action





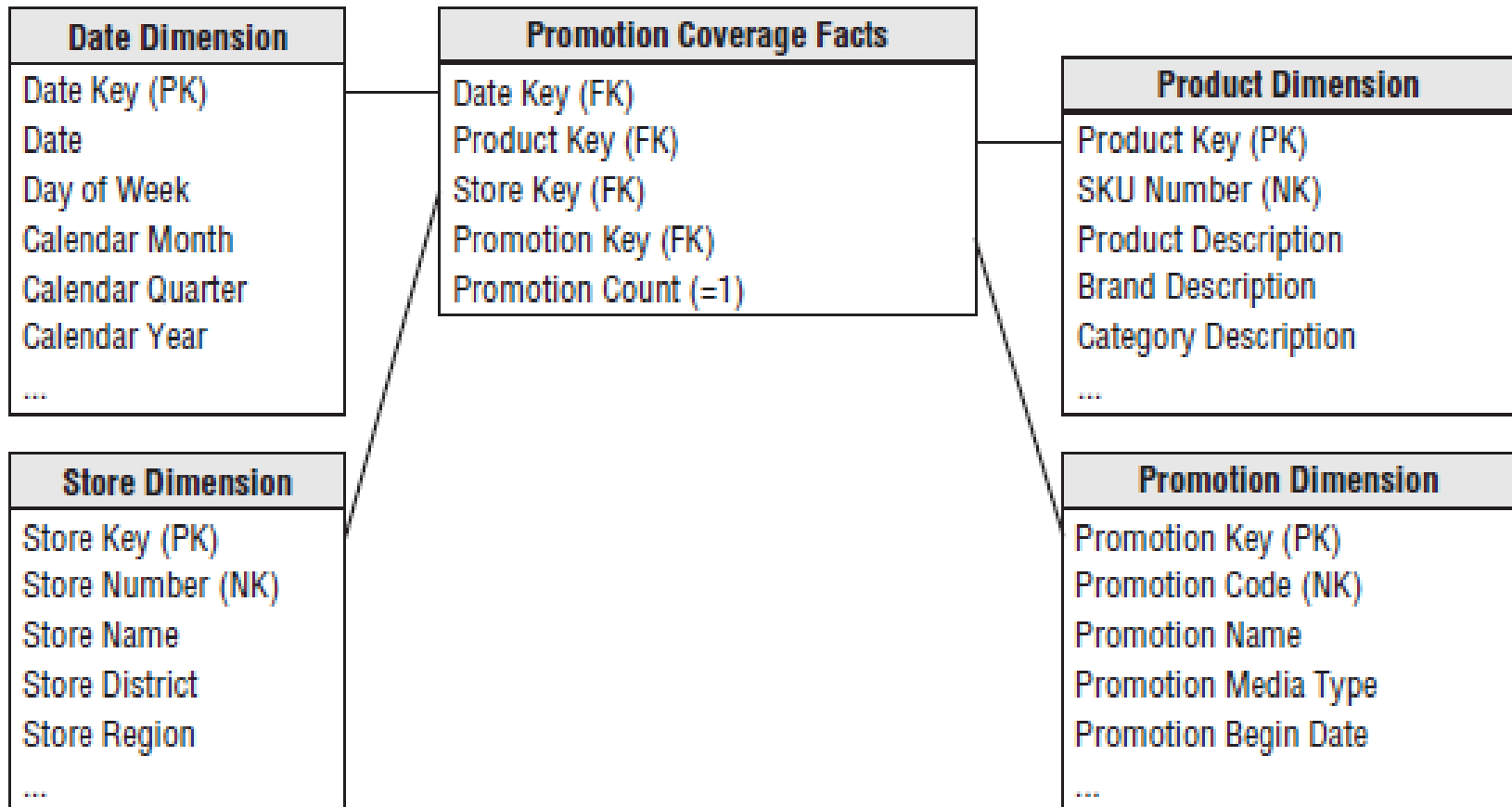
# Retail Schema Extensibility

- New dimension attributes
- New dimensions
- New measured facts

# Factless Fact Tables

- What products were on promotion but did not sell?  
The sales fact table records only the SKUs actually sold. How to know what did not happen !
- The promotion coverage factless table **keys** would be **date**, **product**, **store**, and **promotion** in this case study
- Factless table has **zero** facts

# Factless Fact Tables (Cont.)



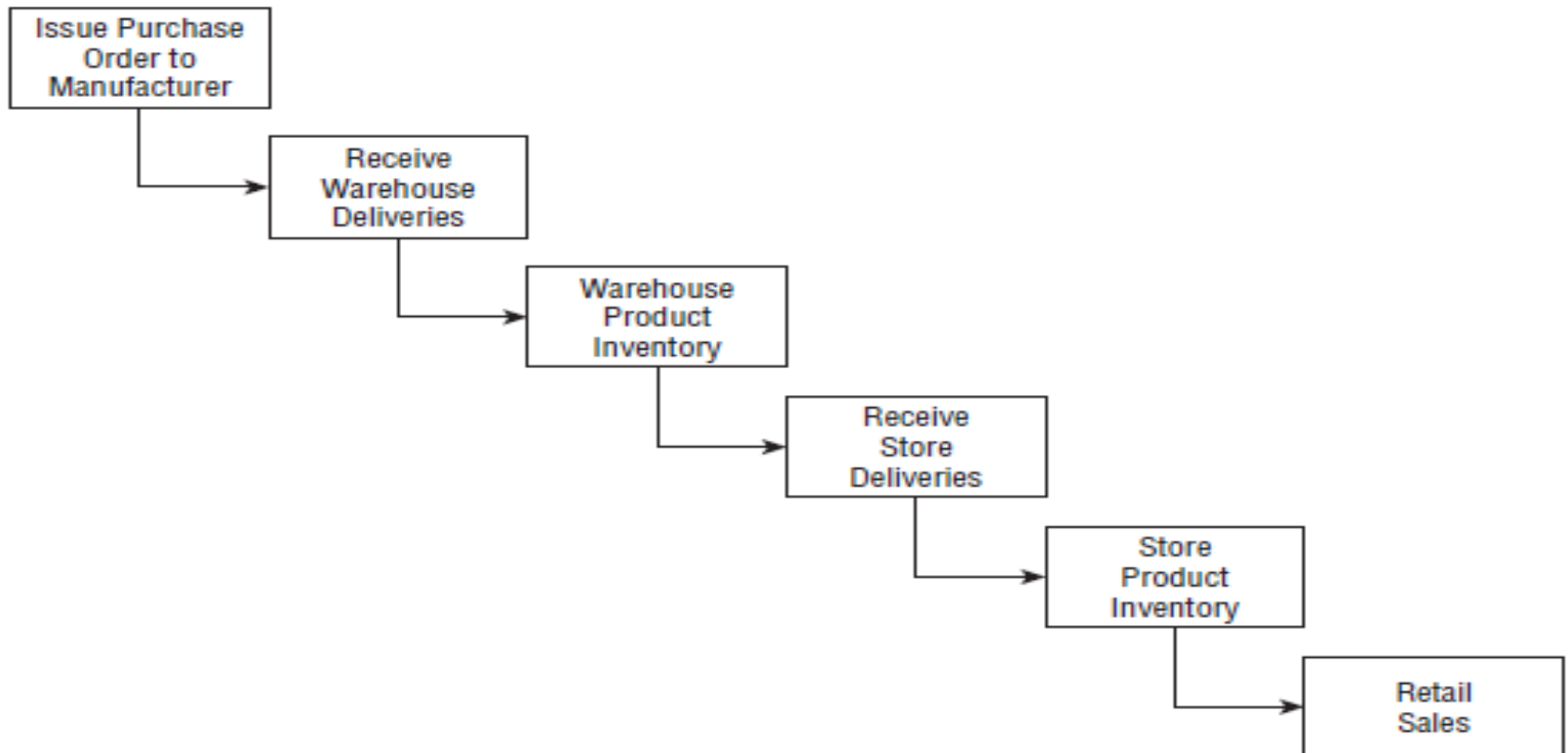


## Chapter 4

# Dimensional Modeling (Store Inventory/ Warehouse Product Inventory)

# Value Chain

- The value chain identifies the natural, logical flow of an organization's primary activities



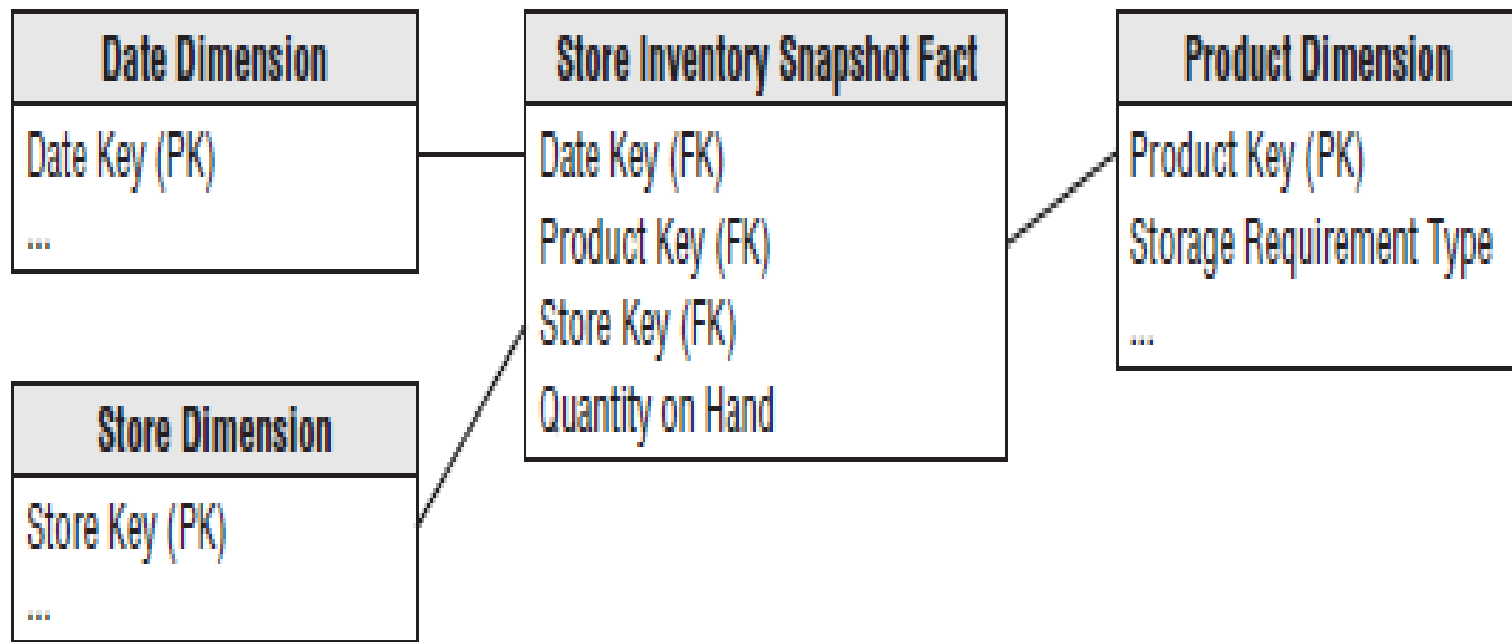
# Inventory Models

- Store Inventory Periodic Snapshot
- Product Warehouse Accumulating Snapshot

# Store Inventory Periodic Snapshot

- **Objective:** Making sure the right product is in the right store at the **right time minimizes out-of-stocks** (where the product isn't available on the shelf to be sold)
- **Dimensional Design Steps**
  1. Business Process: analyzing is the periodic snapshotting of retail store inventory
  2. Grain: atomic level , daily inventory
  3. Dimensions: date, product and store
  4. Facts: quantity on hand

# Store Inventory Periodic Snapshot



- Dense or Sparse table ?
- Quantity on Hand is **semi-additive** fact



# Product Warehouse Accumulating Fact

- Objective: used for processes that have a definite beginning, end and identifiable milestones in between.
- Product Warehouse Transaction Process
  - ❖ Receive product from manufacturer
  - ❖ Place product into inspection hold.
  - ❖ Release product from inspection hold.
  - ❖ Return product to vendor due to inspection failure.
  - ❖ Place product in bin.
  - ❖ Pick product from bin.
  - ❖ Package product for shipment.
  - ❖ Ship product to customer.

# Product Warehouse Accumulating Fact

Fact row inserted when lot received:

| Lot Receipt Number | Date Received Key | Date Inspected Key | Date Bin Placement Key | Product Key | Quantity Received | Receipt to Inspected Lag | Receipt to Bin Placement Lag |
|--------------------|-------------------|--------------------|------------------------|-------------|-------------------|--------------------------|------------------------------|
| 101                | 20130101          | 0                  | 0                      | 1           | 100               |                          |                              |

Fact row updated when lot inspected:

| Lot Receipt Number | Date Received Key | Date Inspected Key | Date Bin Placement Key | Product Key | Quantity Received | Receipt to Inspected Lag | Receipt to Bin Placement Lag |
|--------------------|-------------------|--------------------|------------------------|-------------|-------------------|--------------------------|------------------------------|
| 101                | 20130101          | 20130103           | 0                      | 1           | 100               | 2                        |                              |

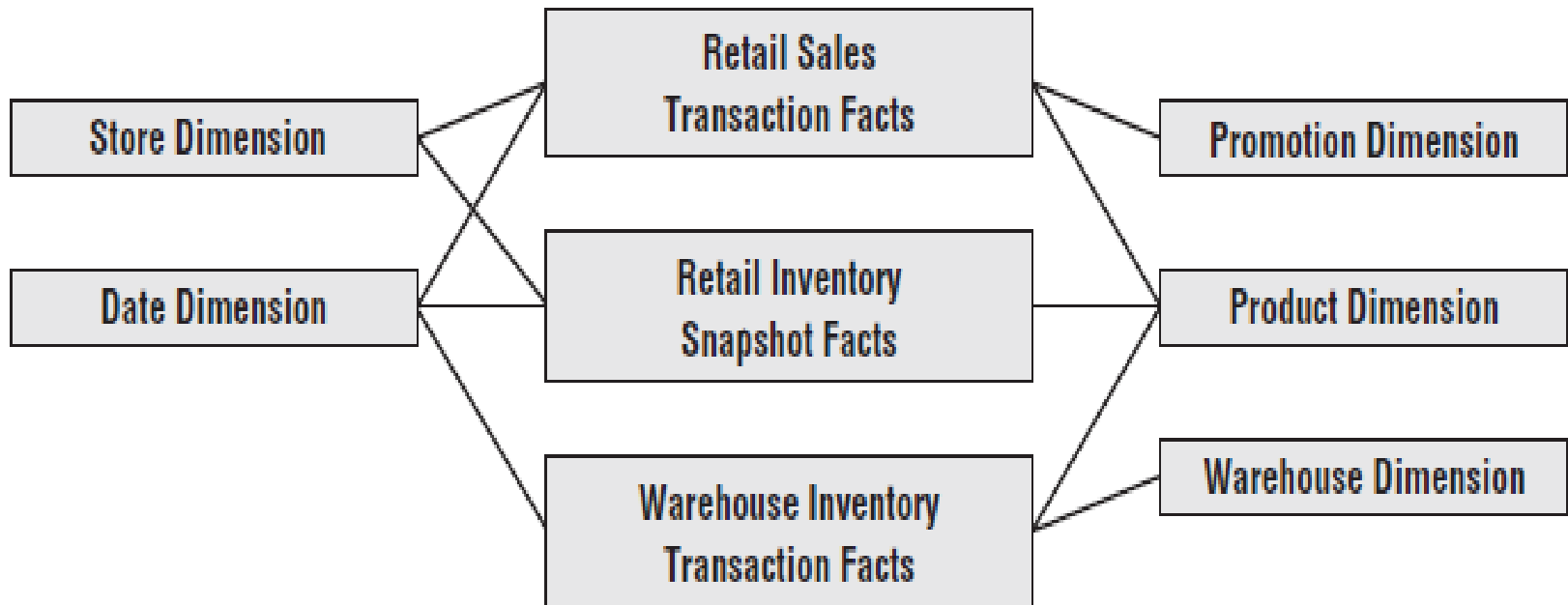
Fact row updated when lot placed in bin:

| Lot Receipt Number | Date Received Key | Date Inspected Key | Date Bin Placement Key | Product Key | Quantity Received | Receipt to Inspected Lag | Receipt to Bin Placement Lag |
|--------------------|-------------------|--------------------|------------------------|-------------|-------------------|--------------------------|------------------------------|
| 101                | 20130101          | 20130103           | 20130104               | 1           | 100               | 2                        | 3                            |

# Fact Table Types

|                     | Transaction                               | Periodic Snapshot                                     | Accumulating Snapshot                                  |
|---------------------|-------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|
| Periodicity         | Discrete transaction point in time        | Recurring snapshots at regular, predictable intervals | Indeterminate time span for evolving pipeline/workflow |
| Grain               | 1 row per transaction or transaction line | 1 row per snapshot period plus other dimensions       | 1 row per pipeline occurrence                          |
| Date dimension(s)   | Transaction date                          | Snapshot date                                         | Multiple dates for pipeline's key milestones           |
| Facts               | Transaction performance                   | Cumulative performance for time interval              | Performance for pipeline occurrence                    |
| Fact table sparsity | Sparse or dense, depending on activity    | Predictably dense                                     | Sparse or dense, depending on pipeline occurrence      |
| Fact table updates  | No updates, unless error correction       | No updates, unless error correction                   | Updated whenever pipeline activity occurs              |

# Value Chain Integration



- Fact Constellation/ Galaxy Schema
- Conformed Dimension (Shared dimension, common dimension)

# Slowly Changing Dimension

- Dimension table attributes are relatively static; but are not fixed forever
- Dimensions attributes might change slowly over time
- Business Users should determine if they are interested to track dimension attributes change or not !
- Some dimension attributes values never change. It applies for any attribute labeled *original* or *initial*  
Ex: customer initial balance  
(Type o : Retain Original)

# SCD Types

- Type 1: Overwrite
  - Overwriting preexisting value with the current assignment
  - Type 1 is easy to implement but it does not maintain any history of prior attribute values
  - No changes required for the fact table

Original row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name |
|-------------|----------|---------------------|-----------------|
| 12345       | ABC922-Z | IntelliKidz         | Education       |

Updated row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name |
|-------------|----------|---------------------|-----------------|
| 12345       | ABC922-Z | IntelliKidz         | Strategy        |

# SCD Types

- Type 2: Add New Row
  - Adding a new row to indicate previous and current values
  - Type 2 is used to partition or segment history to consider the changes
  - Fact table shall be updated by the new product key for new product

Original row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|-------------|----------|---------------------|-----------------|-----|--------------------|---------------------|-----------------------|
| 12345       | ABC922-Z | IntelliKidz         | Education       | ... | 2012-01-01         | 9999-12-31          | Current               |

Rows in Product dimension following department reassignment:

| Product Key | SKU (NK) | Product Description | Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|-------------|----------|---------------------|-----------------|-----|--------------------|---------------------|-----------------------|
| 12345       | ABC922-Z | IntelliKidz         | Education       | ... | 2012-01-01         | 2013-01-31          | Expired               |
| 25984       | ABC922-Z | IntelliKidz         | Strategy        | ... | 2013-02-01         | 9999-12-31          | Current               |

# SCD Types

- Type 3: Add New Attribute
  - Adding a new attribute to indicate previous value
  - Type 3 enables you to see new and historical fact data
  - Fact table is unaffected
  - Not frequently used; fixed number of changes

Original row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name |
|-------------|----------|---------------------|-----------------|
| 12345       | ABC922-Z | IntelliKidz         | Education       |

Updated row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name | Prior Department Name |
|-------------|----------|---------------------|-----------------|-----------------------|
| 12345       | ABC922-Z | IntelliKidz         | Strategy        | Education             |



# DWH Data Models

- Star Schema
- Snowflake Schema
- 3NF Schema



# Chapter 5

## Physical Design

# Defining the Physical Model

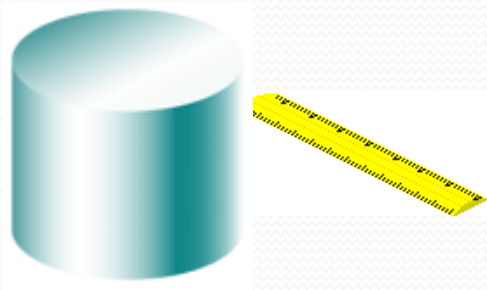
- Translate the dimensional design to a physical model for implementation.
- Determine the hardware architecture.
- Define the storage strategy for tables and indexes.
- Perform database sizing.
- Define the partitioning strategy.
- Define the initial indexing strategy.
- Define the security strategy.

# Translating a Dimensional Model to a Physical Model

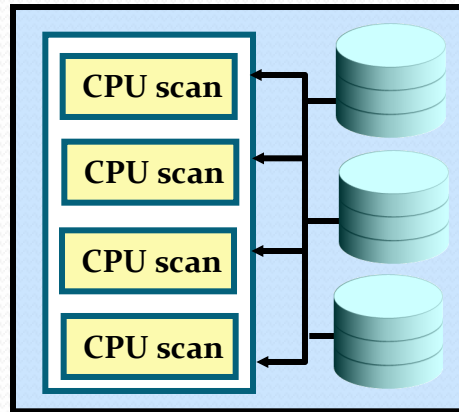
- Develop object naming conventions.
- Apply the naming standards to the tables and attributes of the dimensional model.

|                      |        |
|----------------------|--------|
| * PRODUCT_ID         | v(11)  |
| * PROD_DESC          | v(125) |
| * PROD_NAME          | v(35)  |
| * PROD_CATEGORY_ID   | v(20)  |
| * PROD_CATEGORY_DESC | v(50)  |
| * PROD_SUBCATEGORY   | v(25)  |
| * SUPPLIER_ID        | v(20)  |
| * PROD_STATUS        | v(10)  |
| * PROD_LIST_PRICE    | n      |
| * PROD_MIN_PRICE     | v(20)  |
| * PROD_PACK_SIZE     | v(20)  |
| * PROD_WEIGHT_CLASS  | v(10)  |
| * PROMOTION_CODE     | v(10)  |
| * WHSE_LOCATION      | v(10)  |
| * PROD_EFF_FROM      | date   |
| * PROD_EFF_TO        | date   |

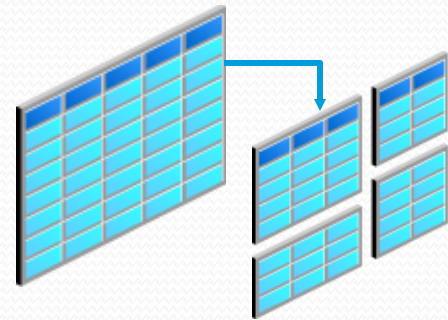
# Storage and Performance Considerations



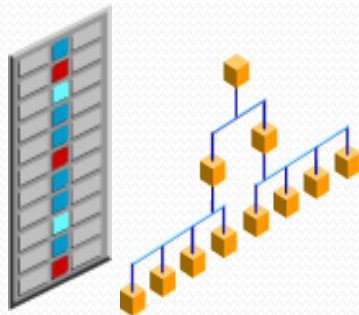
**Database Sizing**



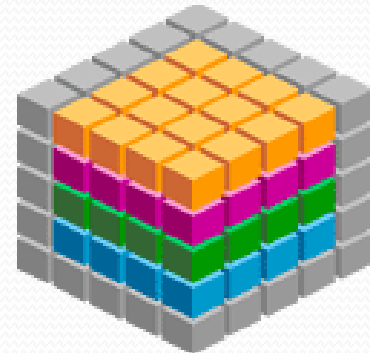
**Parallelism**



**Data Partitioning**



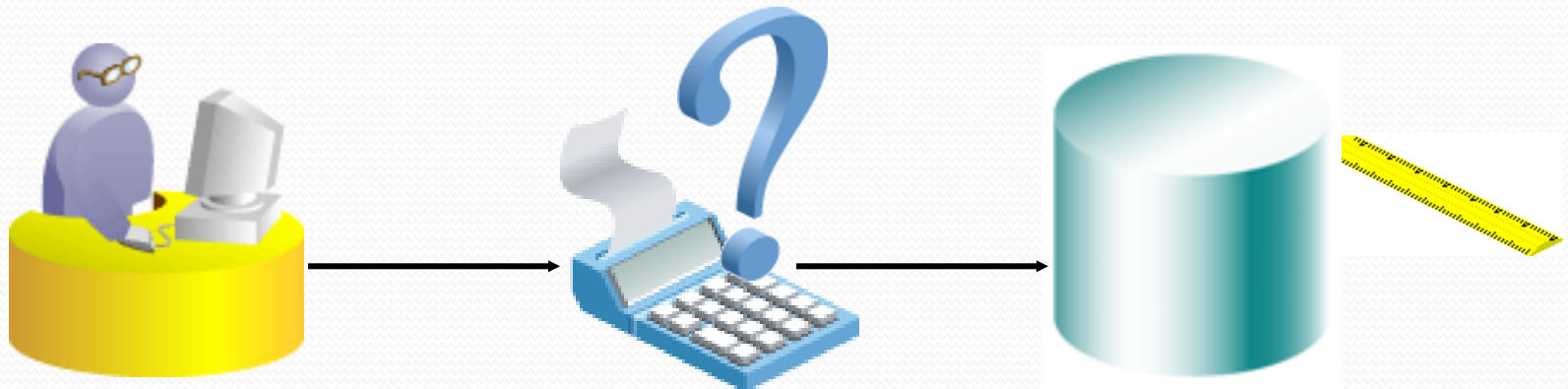
**Indexing**



**Summary Data**

# Estimating the Database Size

1. Estimate the size of each row in the fact table.
2. Determine the grain of each dimension and estimate the number of entries in the finest level.
3. Multiply the number of rows of all dimensions and multiply the result by the fact table row size.
4. Determine whether the fact table is sparse or dense and estimate the reduction or increase in size.



# Validating Database Size Assumptions

| Description                                                                                                                                    | Estimation                                                                         |                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1. Estimate the size of one row of the fact table.                                                                                             | 52 bytes (assumed for this example)                                                |                                                            |
| 2. Determine the grain of each dimension and estimate the number of entries in the finest level.                                               | <b>Stores</b><br><b>Location</b><br><b>Products</b><br><b>History</b>              | 5 Stores<br>Located in 23 countries<br>72 items<br>48 Days |
| 3. Multiply the number of entries for each dimension and multiply the result by the fact table row size.                                       | $(5 \times 23 \times 72 \times 48) \times 52 = 20,666,880$ bytes                   |                                                            |
| 4. Determine whether the fact table is sparse or dense and estimate the reduction or increase in size; i.e. if sparsity is low; adjust by 10%. | $20,666,880 \times 0.1 = 2,066,688$<br>$20,666,880 - 2,066,688 = 18,600,192$ bytes |                                                            |
| <b>Estimated database size</b>                                                                                                                 | <b>18.6 MB (approximately)</b>                                                     |                                                            |

# Testing Load Sampling

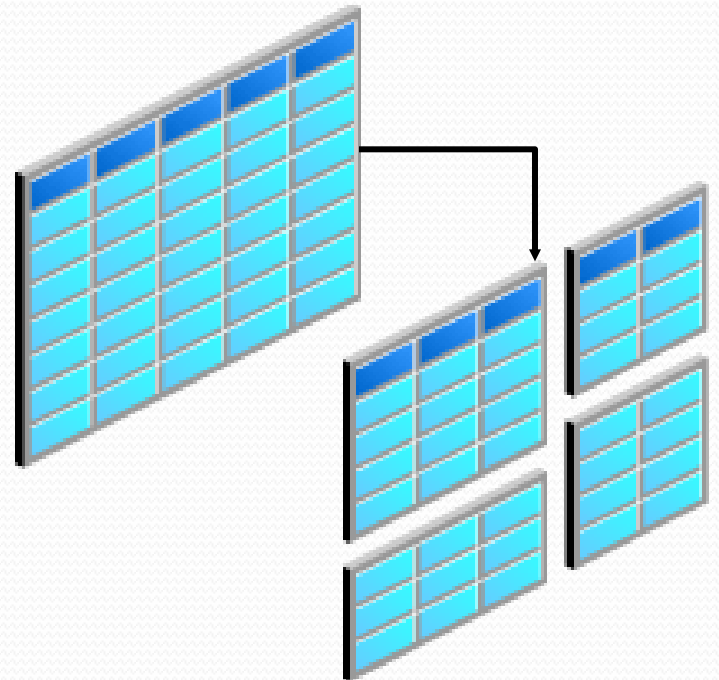
- Analyze a representative sample of the data chosen using proven statistical methods. Ensure that the sample reflects:
  - Test loads for different periods
  - Day-to-day operations
  - Seasonal data and worst-case scenarios





# Data Partitioning

- Breaking up of data into smaller units that can be handled independently
- Data partitioning provides ease of:
  - Restructuring
  - Reorganization
  - Removal
  - Recovery
  - Monitoring
  - Management
  - Archiving



# Why Data Partitioning Is Needed

- Data warehouses may:
  - Grow to become very large databases (VLDB)
  - Have very huge fact tables and lots of historical data
- Table availability:
  - Large tables are more vulnerable to disk failure.
  - It is too costly to have a large table inaccessible for hours due to recovery.
- Large table manageability:
  - Partial deletes take hours, even days.
- Performance considerations:
  - Large table scans are costly.
  - Scanning a subset improves performance.

# Partition Performance Benefits: Partition Pruning

**SALES**

01-Jan

01-Feb

01-Mar

01-Apr

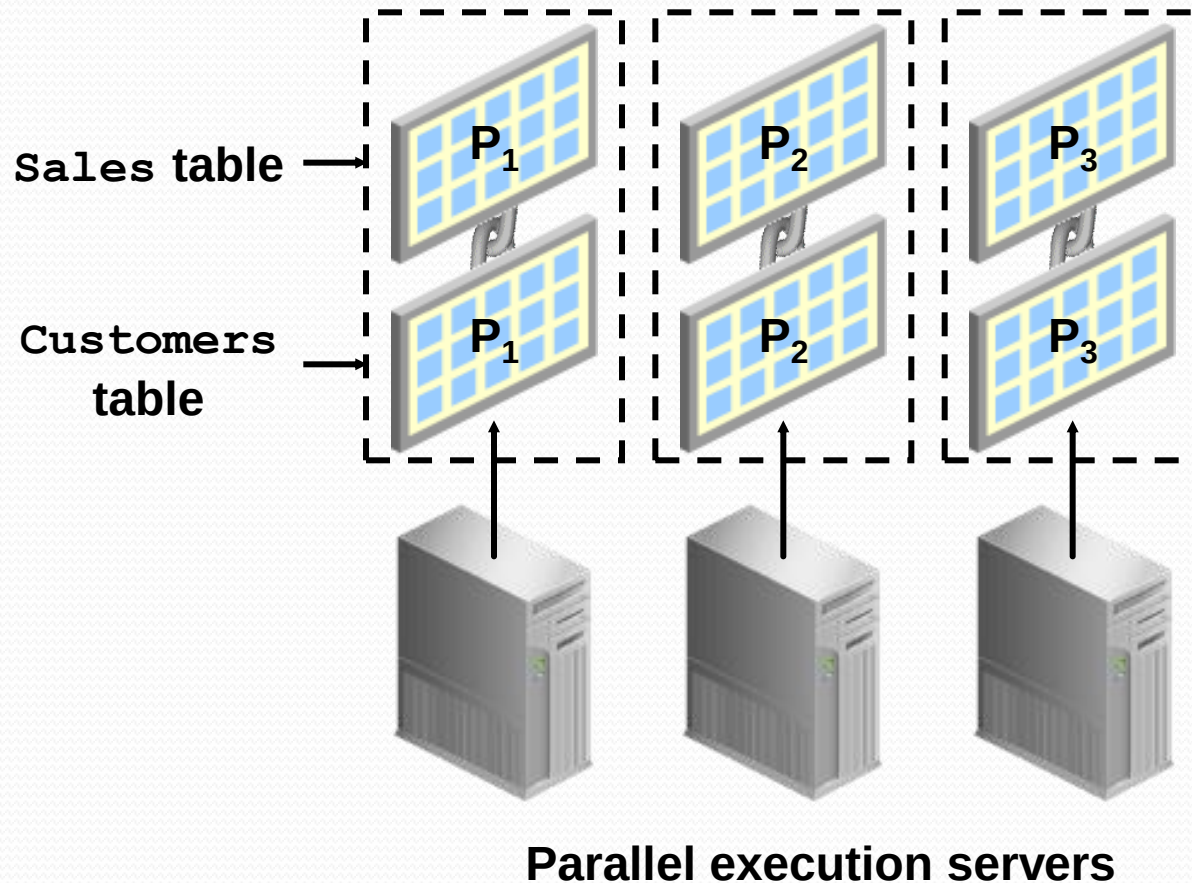
01-May

01-Jun

**Partition pruning:  
Only the relevant partitions are accessed.**

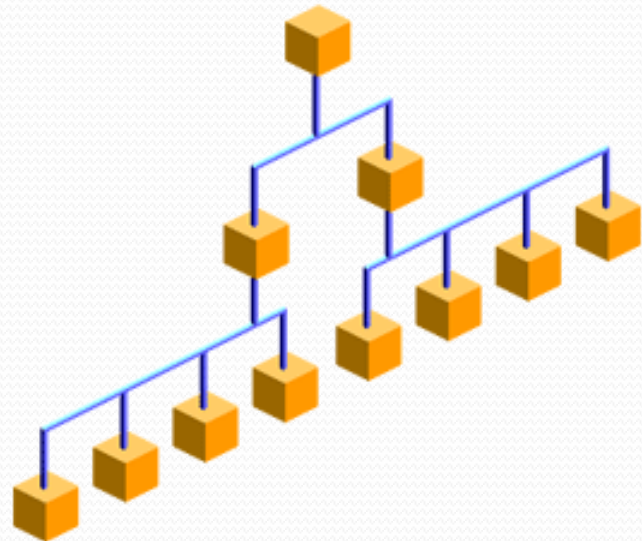
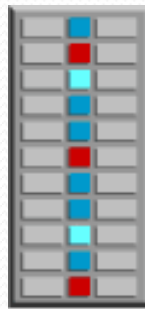
```
SELECT SUM(amount_sold)
FROM sales
WHERE time_id BETWEEN
      TO_DATE('01-MAR-2000',
              'DD-MON-YYYY') AND
      TO_DATE('31-MAY-2000',
              'DD-MON-YYYY');
```

# Parallelism



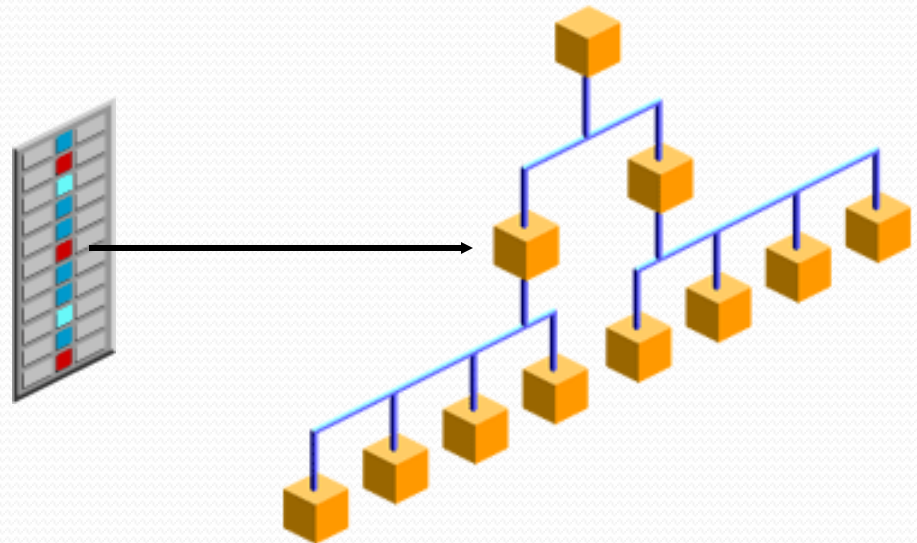
# Indexing

- Indexing is used for the following reasons:
  - It saves cost and greatly improves performance and scalability.
  - It can replace a full-table scan by a quick read of the index followed by a read of only those disk blocks that contain the rows needed.



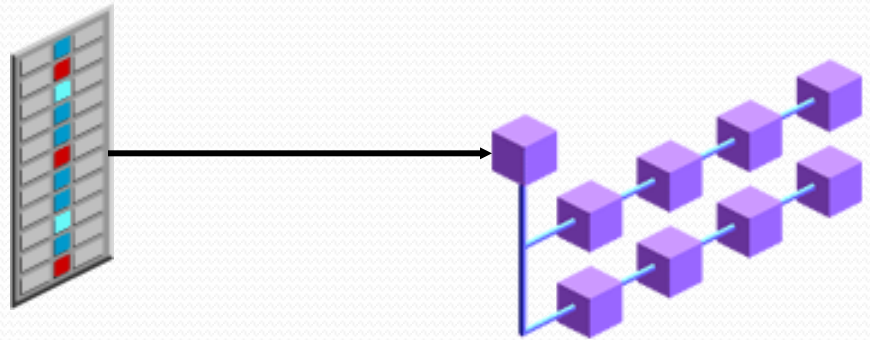
# B-Tree Index

- Most common type of indexing
- Used for high-cardinality columns
- Designed for few rows returned



# Bitmap Indexes

- Provide performance benefits and storage savings
- Store values as 1s and 0s
- Can be used instead of B-tree indexes when:
  - Tables are large
  - Columns have low cardinality



# Bitmap Index: Example

Customer table

| CUSTOMER_No | MARITAL_STATUS | REGION  | GENDER | INCOME_LEVEL |
|-------------|----------------|---------|--------|--------------|
| 101         | single         | east    | male   | bracket_1    |
| 102         | married        | central | female | bracket_4    |
| 103         | married        | west    | female | bracket_2    |
| 104         | divorced       | west    | male   | bracket_4    |
| 105         | single         | central | female | bracket_2    |
| 106         | married        | central | female | bracket_3    |

Sample bitmap index on the REGION column

```
CREATE BITMAP INDEX REGION_IDX ON CUSTOMER(REGION) ;
```

| REGION='east' | REGION = 'central' | REGION = 'west' |
|---------------|--------------------|-----------------|
| 1             | 0                  | 0               |
| 0             | 1                  | 0               |
| 0             | 0                  | 1               |
| 0             | 0                  | 1               |
| 0             | 1                  | 0               |
| 0             | 1                  | 0               |



# Bitmap Index: Example

```
SELECT COUNT(*)  
FROM CUSTOMER  
WHERE MARITAL_STATUS = 'married' AND  
REGION IN ('central', 'west');
```

| MARITAL_STATUS =<br>'married' |     | REGION = 'central'<br>OR<br>REGION = 'west' |    |   | Query<br>Result |
|-------------------------------|-----|---------------------------------------------|----|---|-----------------|
| 0                             |     | 0                                           |    | 0 | 0               |
| 1                             |     | 1                                           |    | 0 | 1               |
| 1                             | AND | 0                                           | OR | 1 | 1               |
| 0                             |     | 0                                           |    | 1 | 0               |
| 0                             |     | 1                                           |    | 0 | 0               |
| 1                             |     | 1                                           |    | 0 | 1               |

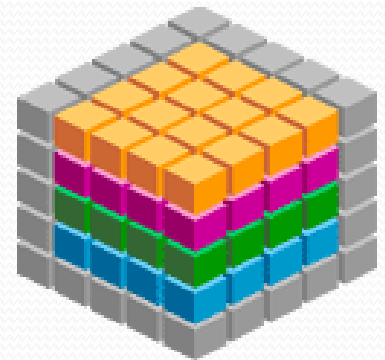
col 2

col 3

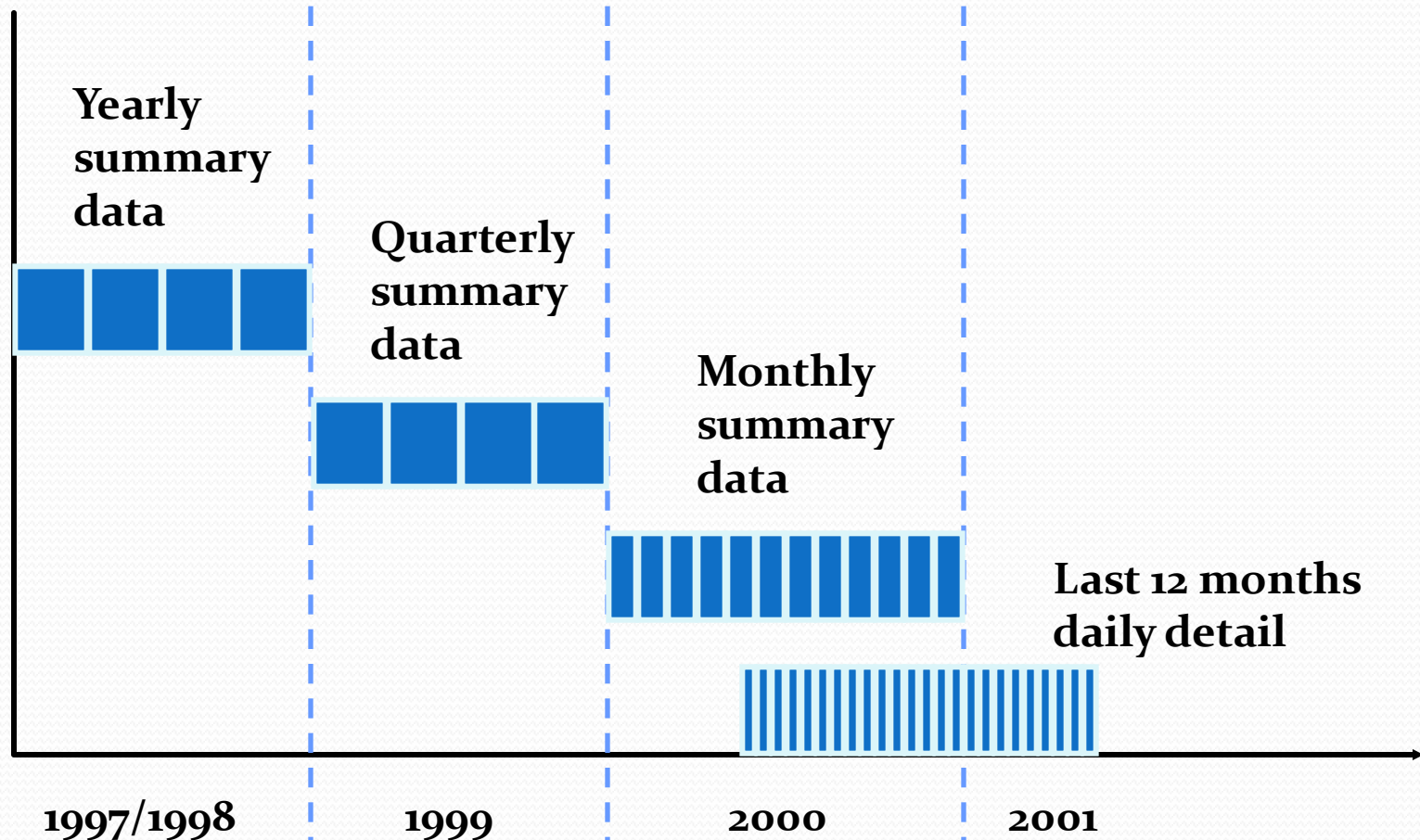
col 5

# Using Summary Data

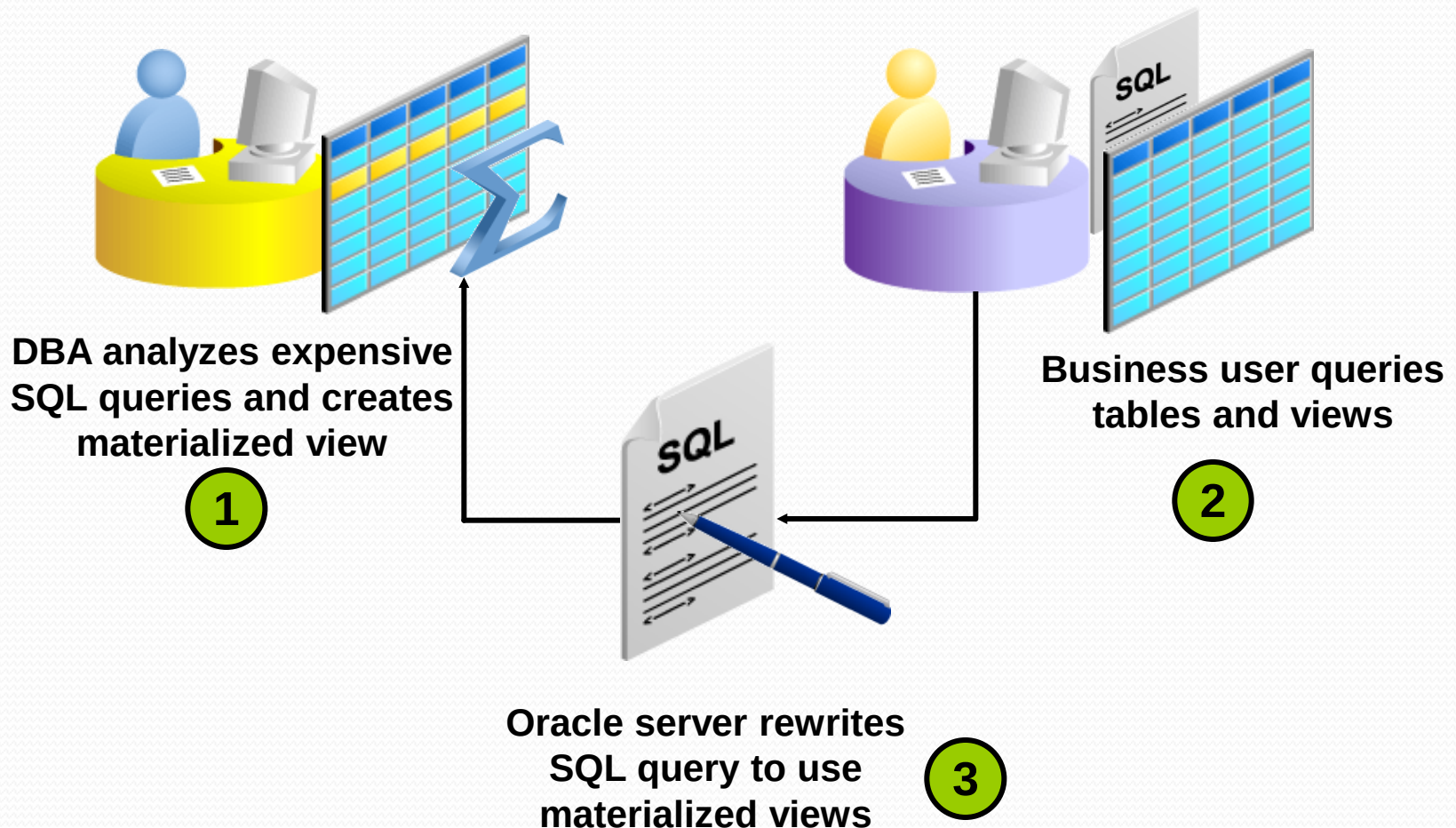
- Designing summaries offers the following benefits:
  - Provides fast access to precomputed data
  - Reduces use of I/O, CPU, and memory
- Materialized View
  - Store precomputed aggregates and joins.
  - Results are stored in the database.
  - Use query rewrite.
  - Improve query performance.



# Managing Historical Summary Data in the Warehouse

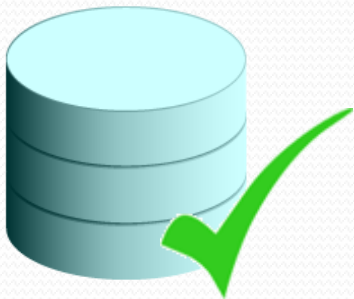


# Using Materialized Views for Summary Management

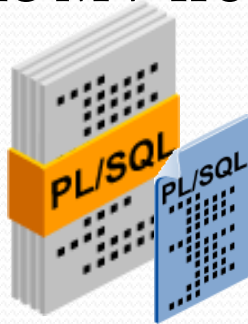


# Available Refresh Modes When Creating a Materialized View

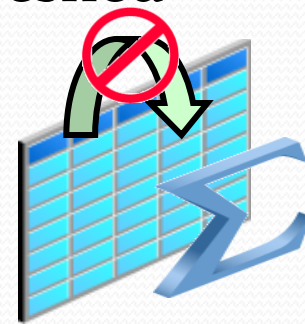
- **ON DEMAND:** Enables manual refresh using procedures in the `DBMS_MVIEW` package (default)
- **ON COMMIT:** Enables refresh at transaction commit
  - Possible only for fast-refreshable materialized views
  - The `ON COMMIT REFRESH` object privilege needed
  - Manual subsequent refreshes in case of failure
- **NEVER:** Prevents the MV from being refreshed



ON COMMIT



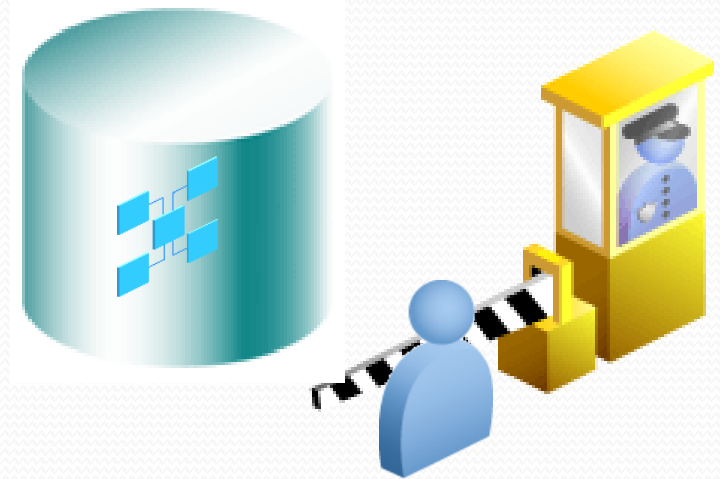
ON DEMAND using the  
`DBMS_MVIEW` package



NEVER

# Security in Data Warehouses

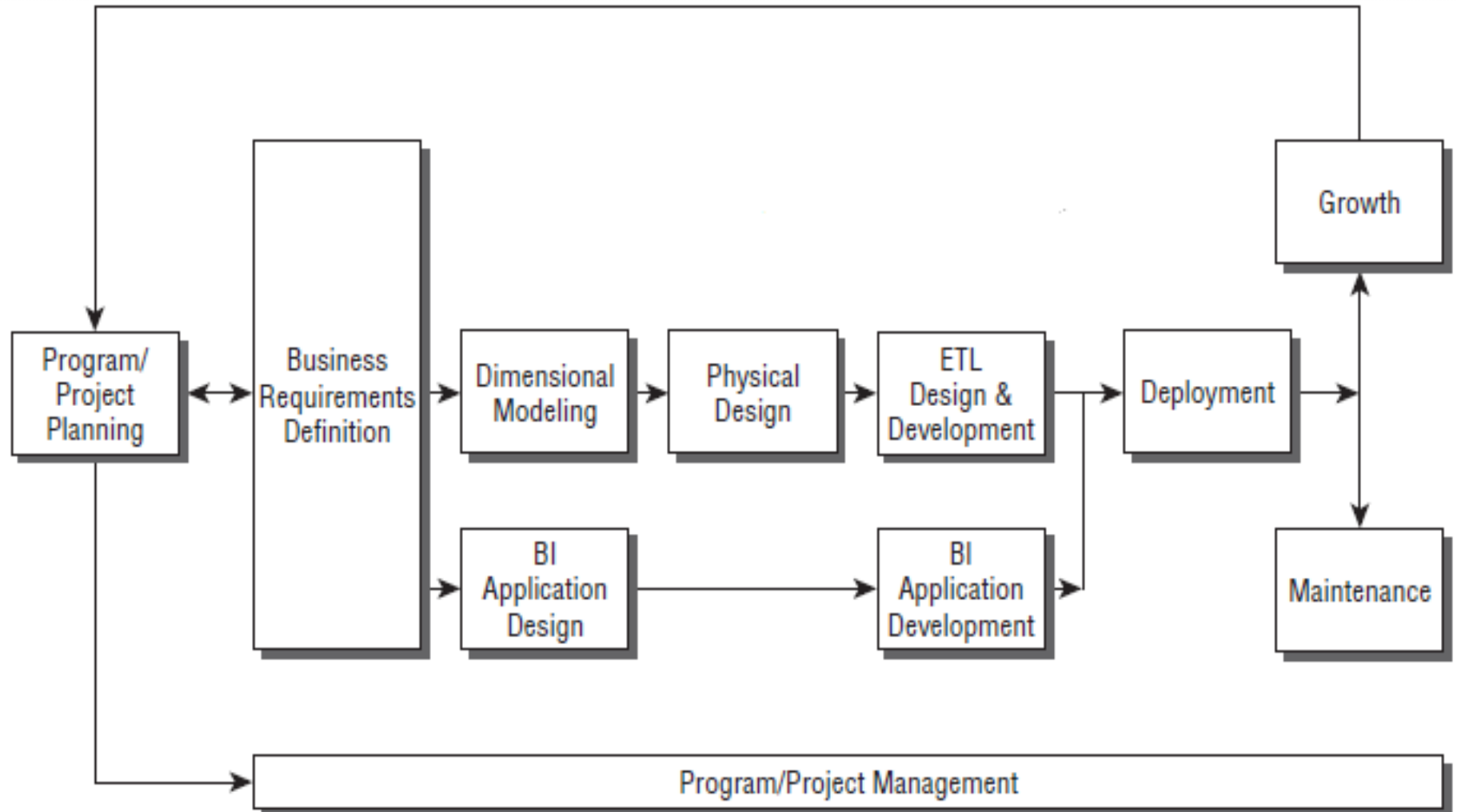
- Why is security important in data warehouses?
  - To prevent unauthorized access
  - To avoid data theft by hackers
  - To provide the right data to the right set of users
  - To keep a record of the user activities



# DW Vendors in Market !



# DW/BI Project Life Cycle





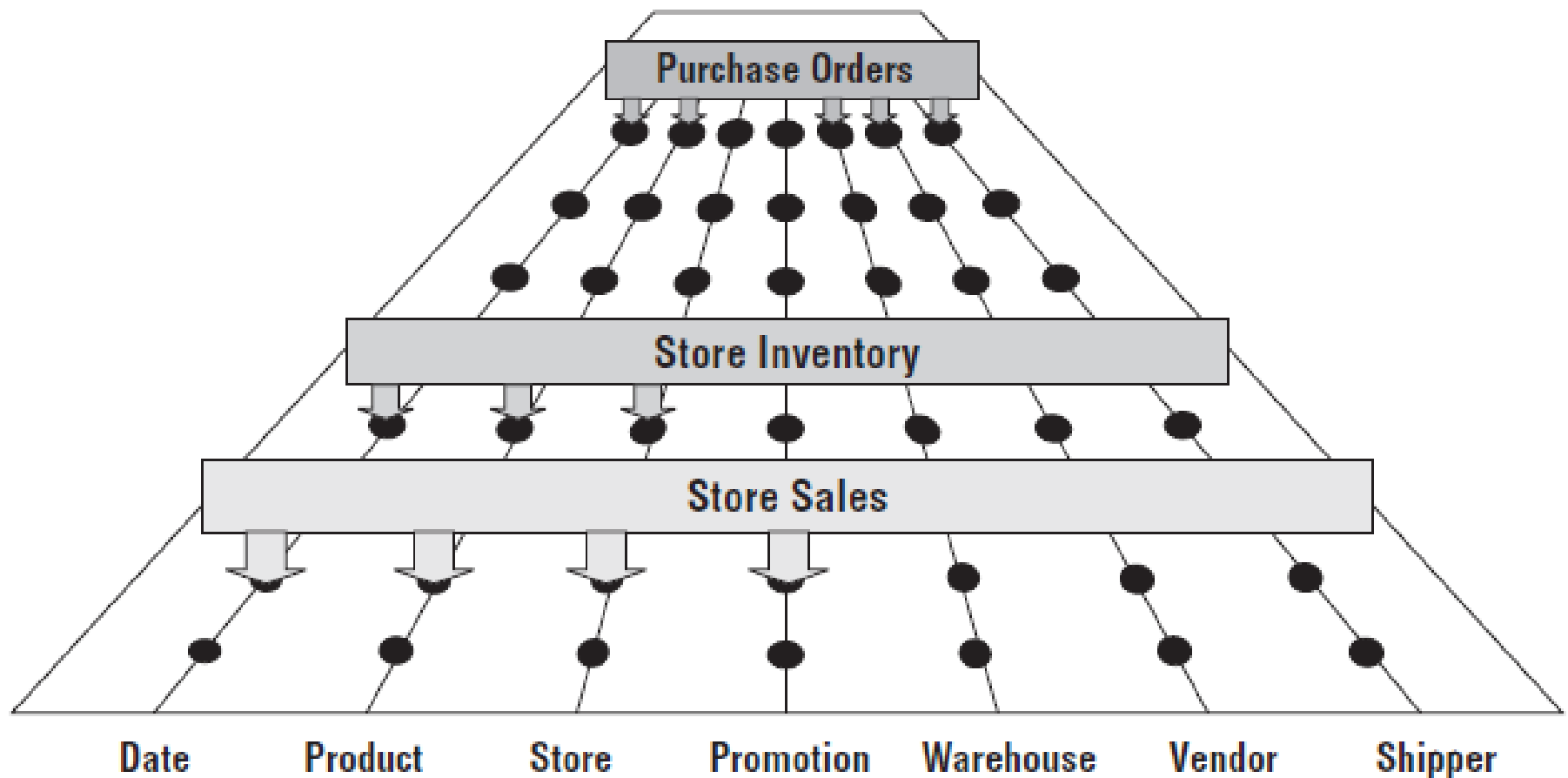
# BI Governance

***Definition:*** The process which is responsible for project prioritization.

## ***BI Governance Tasks:***

1. Create categories of projects (investment, business opportunity, strategic, mandatory, etc.)
2. Define criteria for project selection
3. Determine and set a framework for managing project risk
4. Manage and leverage project interdependencies

# Enterprise DWH Bus Architecture



# Enterprise DWH Bus Matrix

| BUSINESS PROCESSES           | COMMON DIMENSIONS |         |           |       |           |          |          |
|------------------------------|-------------------|---------|-----------|-------|-----------|----------|----------|
|                              | Date              | Product | Warehouse | Store | Promotion | Customer | Employee |
| Issue Purchase Orders        | X                 | X       | X         |       |           |          |          |
| Receive Warehouse Deliveries | X                 | X       | X         |       |           |          | X        |
| Warehouse Inventory          | X                 | X       | X         |       |           |          |          |
| Receive Store Deliveries     | X                 | X       | X         | X     |           |          | X        |
| Store Inventory              | X                 | X       |           | X     |           |          |          |
| Retail Sales                 | X                 | X       |           | X     | X         | X        | X        |
| Retail Sales Forecast        | X                 | X       |           | X     |           |          |          |
| Retail Promotion Tracking    | X                 | X       |           | X     | X         |          |          |
| Customer Returns             | X                 | X       |           | X     | X         | X        | X        |
| Returns to Vendor            | X                 | X       |           | X     |           |          | X        |
| Frequent Shopper Sign-Ups    | X                 |         |           | X     |           | X        | X        |

# Opportunity/ Stakeholders Matrix

| BUSINESS PROCESSES           | STAKEHOLDERS  |           |                  |           |         |
|------------------------------|---------------|-----------|------------------|-----------|---------|
|                              | Merchandising | Marketing | Store Operations | Logistics | Finance |
| Issue Purchase Orders        | X             |           | X                | X         | X       |
| Receive Warehouse Deliveries | X             |           | X                | X         | X       |
| Warehouse Inventory          | X             |           | X                | X         | X       |
| Receive Store Deliveries     | X             |           | X                | X         | X       |
| Store Inventory              | X             | X         | X                | X         | X       |
| Retail Sales                 | X             | X         | X                | X         | X       |
| Retail Sales Forecast        | X             | X         | X                | X         | X       |
| Retail Promotion Tracking    | X             | X         | X                | X         | X       |
| Customer Returns             | X             |           | X                | X         | X       |
| Returns to Vendor            | X             |           | X                | X         | X       |
| Frequent Shopper Sign-Ups    |               | X         | X                |           | X       |

# DWH Projects Examples

- Reporting Requirements (Reports, Dashboards, Score Cards, ...)
- Pre Assessments / Post Assessments (Adhoc Requests)
- Business Models/Presentation Layer (Ex: Inactivity Model, Customer Segment Model, Migration Model, etc)
- Interfaces



Thank you !